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Surgical systems and devices, including methods for configuring surgical systems and devices, for managing postpartum hemorrhaging

Abstract

Embodiments relate to a hysteroscopic system for managing post-partum hemorrhaging. The hysteroscopic system includes a main assembly and a uterine assembly. The main assembly includes a main channel, a cervical seal member, a first anchoring member, a second anchoring member, and a third anchoring member. The uterine assembly includes a first negative pressure port, a uterine expandable member, and a second negative pressure port.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to systems, devices, and methods for managing postpartum hemorrhaging, and in particular, to surgical systems, subsystems, processors, devices, logic, methods, and processes for managing and controlling postpartum hemorrhaging.

BACKGROUND

(2) Postpartum hemorrhaging (or “PPH”) is the leading cause of maternal morbidity and mortality worldwide, responsible for more than 25% of deaths annually. PPH is generally understood and accepted as blood loss of greater than 500 ml for vaginal delivery and greater than 1,000 ml for caesarean delivery, although accurate estimation of blood loss at delivery and subsequent ascertainment of PPH can be challenging. Leading causes of PPH include uterine atony (failure of the uterus to contract), lacerations, retained placenta or clots, and clotting factor deficiency secondary to massive blood transfusion. In addition to death, other serious morbidity resulting from postpartum bleeding may include multi-organ failure, shock, adult respiratory distress syndrome, disseminated coagulopathy, and loss of fertility due to emergency hysterectomy (i.e., removal of the uterus).

(3) Most cases of PPH have no identifiable risk factors. Despite advances in the field of obstetrics, PPH remains one of the main causes of maternal death in several countries. In fact, approximately 14 million women suffer from PPH, and of these, it is estimated that more than 125,000 women die of PPH in the world annually. Likewise, it is well known that 99% of these deaths occur in developing countries. Generally, PPH-related symptoms include, but are not limited to, dizziness, loss of consciousness, presence of hypotension, acceleration of heart rate, and an abnormal reduction in urine. These symptoms may last up to twenty-four hours (or more or less).

(4) Uterine contraction is a natural physiological process that controls bleeding by constricting the vasculature of the uterus. If abnormal postpartum uterine bleeding occurs, initial conservative management, including uterine massage and administration of therapeutic oxytocin, is performed.

If these interventions are found to be unsuccessful in regaining uterine tone and controlling uterine bleeding, additional conservative methods such as therapeutic doses of uterotonic medications are implemented. If these medications do not adequately control abnormal postpartum uterine bleeding or cannot be used in a patient due to contraindications, additional interventions are required.

(5) However, the above methods usually take long time to completely control the PPH. Whereas immediately after diagnosis of primary PPH, it is imperative to implement appropriate and timely therapeutic measures as most deaths occur within the first few hours after onset of major bleeding. As such, an immediate arrest of bleeding is essential to save life.

BRIEF SUMMARY

(6) Present example embodiments relate generally to and/or include systems, subsystems, assemblies, processors, and devices for addressing conventional problems, including those described above and in the present disclosure, and more specifically, example embodiments relate to systems, subsystems, assemblies, processors, and devices for managing post-partum hemorrhaging.

(7) In an exemplary embodiment, a hysteroscopic system for managing post-partum hemorrhaging is described. The hysteroscopic system includes a main assembly. The main assembly is formed as an elongated body having an in vivo end and an in vitro end. When in operation, the in vivo end of the main assembly is configured to be inserted into and housed in a vaginal cavity of a patient; and the in vitro end of the main assembly is configured to remain outside of the vaginal cavity of the patient. The main assembly includes a main channel formed through the elongated body of the main assembly between the in vivo and in vitro ends of the main assembly. The main channel is formed along a first central axis (and/or is formed in such a way as to include the first central axis formed through it). The main assembly also includes a cervical seal member. The cervical seal member is formed at the in vivo end of the main assembly. The cervical seal member includes a central axis coaxial to the first central axis. The cervical seal member includes a contact wall and a non-contact wall opposite to the contact wall of the cervical seal member. The contact wall of the cervical seal member is configured in such a way as to face and contact with at least a portion of a cervix of the patient. The cervical seal member is configured to hermetically isolate a uterine cavity of the patient from the vaginal cavity of the patient when the contact wall of the cervical seal member is positioned to be in contact with at least a portion of the cervix of the patient. Alternatively or in addition, the cervical seal member is configured to ensure the main assembly does not enter into the uterine cavity of the patient. The main assembly also includes a first anchoring member. The first anchoring member is formed on (and/or attached to, formed with, etc.) the elongated body of the main assembly and adjacent to the cervical seal member. The first anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the first anchoring member is a state in which the first anchoring member has a first overall volume (e.g., expanded outwardly away from the elongated body of the main assembly towards the wall of the vaginal cavity). The non-anchoring state of the first anchoring member is a state in which the first anchoring member has a second overall volume (e.g., not expanded outwardly away from the elongated body of the main assembly). The first overall volume is greater than the second overall volume. The main assembly also includes a second anchoring member. The second anchoring member is formed on (and/or attached to) the elongated body of the main assembly. The second anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the second anchoring member is a state in which the second anchoring member has a third overall volume (e.g., expanded outwardly away from the elongated body of the main assembly towards the wall of the vaginal cavity). The non-anchoring state of the second anchoring member is a state in which the second anchoring member has a fourth overall volume (e.g., not expanded outwardly away from the elongated body of the main assembly). The third overall volume is greater than the fourth overall volume. The first overall volume may (or may not) be the same volume as the third overall volume. The second

overall volume may (or may not) be the same volume as the fourth overall volume. The main assembly also includes a third anchoring member. The third anchoring member is formed between the first and second anchoring members. The third anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the third anchoring member is a state in which the third anchoring member applies at least a first negative pressure (e.g., suction). The first negative pressure is an amount of negative pressure required to collapse at least a portion of the vaginal cavity (or vaginal wall forming the vaginal cavity) towards the elongated body of the main assembly. The non-anchoring state of the third anchoring member is a state in which the third anchoring member does not apply at least the first negative pressure (e.g., the non-anchoring state of the third anchoring member may be a state in which the third anchoring member does not apply any negative pressure).

(8) The hysteroscopic system also includes a uterine assembly. The uterine assembly is formed as an elongated body having an in vivo end and an in vitro end. The uterine assembly is configured to be slidably housed in the main channel of the main assembly in such a way that the in vivo end of the uterine assembly is extendible outwardly away from the in vivo end of the main assembly (e.g., the uterine assembly can slide into and out of the main channel of the main assembly in both directions). The uterine assembly includes a first negative pressure port. The first negative pressure port is formed at or near a distal end of the in vivo end of the uterine assembly. The first negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the first negative pressure port is a state in which the first negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a second negative pressure. For example, such second negative pressure may be an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly. The non-haemostasis state of the first negative pressure port is a state in which the first negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the second negative pressure (e.g., the non-haemostasis state of the first negative pressure port may be a state in which the first negative pressure port does not apply any negative pressure). The uterine assembly also includes a uterine expandable member. The uterine expandable member is formed on (and/or attached to, formed with, etc.) the elongated body of the uterine assembly and adjacent to the first negative pressure port. The uterine expandable member is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the uterine expandable member is a state in which the uterine expandable member has a third overall volume (e.g., expanded outwardly away from the elongated body of the uterine assembly towards the wall of the uterine cavity). The non-haemostasis state of the uterine expandable member is a state in which the uterine expandable member has a fourth overall volume (e.g., not expanded outwardly away from the elongated body of the uterine assembly). The third overall volume is greater than the fourth overall volume. The uterine assembly also includes a second negative pressure port. The second negative pressure port is formed adjacent to the uterine expandable member in such a way that the uterine expandable member is positioned between the first and second negative pressure ports. The second negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the second negative pressure port is a state in which the second negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a third negative pressure. For example, such third negative pressure may be an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly. The non-anchoring state of the second negative pressure port is a state in which the second negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the third negative pressure (e.g., the non-haemostasis state of the second negative pressure port may be a state in which the second negative pressure port does not apply any negative pressure). The uterine

assembly is configured to apply the second and third negative pressures in such a way that a combination of the second and third negative pressures is a sufficient amount of negative pressure (and/or sufficiently directed in the right directions and/or placed in the right locations within the uterine cavity) so as to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly.

(9) In another exemplary embodiment, a hysteroscopic system for managing post-partum hemorrhaging is described. The hysteroscopic system includes a main assembly. The main assembly is formed as an elongated body having an in vivo end and an in vitro end. The in vivo end of the main assembly is configured to be inserted into and housed in a vaginal cavity of a patient. The in vitro end of the main assembly is configured to remain outside of the vaginal cavity of the patient. The main assembly includes a main channel. The main channel is formed through the elongated body of the main assembly between the in vivo and in vitro ends of the main assembly. The main channel is formed along a first central axis (and/or is formed in such a way as to include the first central axis formed through it). The main assembly also includes a first anchoring member. The first anchoring member is formed on (and/or attached to, formed with, etc.) the elongated body of the main assembly. The first anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the first anchoring member is a state in which the first anchoring member is expanded away from the elongated body of the main assembly (e.g., the anchoring state of the first anchoring member may be a state in which the first anchoring member is protruded towards the vaginal walls forming the vaginal cavity of the patient and/or a state in which the first anchoring member has an increased volume as compared to the non-anchoring state). The main assembly also includes a second anchoring member. The second anchoring member is formed on (and/or attached to) the elongated body of the main assembly. The second anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the second anchoring member is a state in which the second anchoring member is expanded away from the elongated body of the main assembly (e.g., the anchoring state of the second anchoring member may be a state in which the second anchoring member is protruded towards the vaginal walls forming the vaginal cavity of the patient and/or a state in which the second anchoring member has an increased volume as compared to the non-anchoring state). The main assembly also includes a vaginal seal member. The vaginal seal member is formed at the in vitro end of the main assembly. The vaginal seal member may include a central axis coaxial to the first central axis. The vaginal seal member includes a contact wall and a non-contact wall opposite to the contact wall. The contact wall of the vaginal seal member is configured in such a way as to face and contact with at least a portion of a vulva of the patient. The vaginal seal member is configured to hermetically isolate the vaginal cavity of the patient when the contact wall of the vaginal seal member is positioned to be in contact with at least a portion of the vulva of the patient. Alternatively or in addition, the vaginal seal member is configured to ensure the main assembly is properly positioned in the vaginal cavity of the patient. The main assembly also includes a third anchoring member. The third anchoring member is formed between the first and second anchoring members. The third anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the third anchoring member is a state in which the third anchoring member applies at least a first negative pressure (e.g., suction).

(10) The hysteroscopic system also includes a uterine assembly. The uterine assembly is formed as an elongated body having an in vivo end and an in vitro end. The uterine assembly is configured to be slidably housed in the main channel of the main assembly (e.g., the uterine assembly can slide into and out of the main channel of the main assembly in both directions). The uterine assembly includes a first negative pressure port. The first negative pressure port is formed at a distal end of the in vivo end of the uterine assembly. The first negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the first negative pressure port is a state in which the first negative pressure port applies at least a second negative

pressure. For example, such second negative pressure may be an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly. The uterine assembly also includes a uterine expandable member. The uterine expandable member is formed on (and/or attached to, formed with, etc.) the elongated body of the uterine assembly and adjacent to the first negative pressure port. The uterine expandable member is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the uterine expandable member is a state in which the uterine expandable member is expanded away from the elongated body of the uterine assembly (e.g., the haemostasis state of the uterine expandable member may be a state in which the uterine expandable member is protruded towards the uterine wall forming the uterine cavity of the patient and/or a state in which the uterine expandable member has an increased volume as compared to the haemostasis state). The uterine assembly also includes a second negative pressure port. The second negative pressure port is formed adjacent to the uterine expandable member in such a way that the uterine expandable member is positioned between the first and second negative pressure ports. The second negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the second negative pressure port is a state in which the second negative pressure port applies at least a third negative pressure. For example, such third negative pressure may be an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly.
(11) In another exemplary embodiment a hysteroscopic system for managing post-partum hemorrhaging is described. The hysteroscopic system includes a main assembly. The main assembly is formed as an elongated body having an in vivo end and an in vitro end. When in operation, the in vivo end of the main assembly is configured to be inserted into and housed in a vaginal cavity of a patient; and the in vitro end of the main assembly is configured to remain outside of the vaginal cavity of the patient. The main assembly includes a main channel. The main channel is formed through the elongated body of the main assembly between the in vivo and in vitro ends of the main assembly. The main channel is formed along a first central axis (and/or is formed in such a way as to include the first central axis formed through it). The main assembly also includes a cervical seal member. The cervical seal member is formed at the in vivo end of the main assembly. The cervical seal member includes a central axis coaxial to the first central axis. The cervical seal member includes a contact wall and a non-contact wall opposite to the contact wall of the cervical seal member. The contact wall of the cervical seal member is configured in such a way as to face and contact with at least a portion of a cervix of the patient. The cervical seal member is configured to hermetically isolate a uterine cavity of the patient from the vaginal cavity of the patient when the contact wall of the cervical seal member is positioned to be in contact with at least a portion of the cervix of the patient. Alternatively or in addition, the cervical seal member is configured to ensure the main assembly does not enter into the uterine cavity of the patient. The main assembly also includes a first anchoring member. The first anchoring member is formed on (and/or attached to, formed with, etc.) the elongated body of the main assembly and adjacent to the cervical seal member. The first anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the first anchoring member is a state in which the first anchoring member has a first overall volume (e.g., expanded outwardly away from the elongated body of the main assembly towards the wall of the vaginal cavity). The non-anchoring state of the first anchoring member is a state in which the first anchoring member has a second overall volume (e.g., not expanded outwardly away from the elongated body of the main assembly). The first overall volume is greater than the second overall volume. The main assembly also includes a second anchoring member. The second anchoring member is formed on (and/or attached to) the elongated body of the main assembly. The second anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the

second anchoring member is a state in which the second anchoring member has a third overall volume (e.g., expanded outwardly away from the elongated body of the main assembly towards the wall of the vaginal cavity). The non-anchoring state of the second anchoring member is a state in which the second anchoring member has a fourth overall volume (e.g., not expanded outwardly away from the elongated body of the main assembly). The third overall volume is greater than the fourth overall volume. The first overall volume may (or may not) be the same volume as the third overall volume. The second overall volume may (or may not) be the same volume as the fourth overall volume. The main assembly also includes a third anchoring member. The third anchoring member is formed between the first and second anchoring members. The third anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the third anchoring member is a state in which the third anchoring member applies at least a first negative pressure (e.g., suction). The first negative pressure is an amount of negative pressure required to collapse at least a portion of the vaginal cavity (or vaginal wall forming the vaginal cavity of the patient) towards the elongated body of the main assembly. The non-anchoring state of the third anchoring member is a state in which the third anchoring member does not apply at least the first negative pressure (e.g., the non-anchoring state of the third anchoring member may be a state in which the third anchoring member does not apply any negative pressure). The main assembly also includes a vaginal seal member. The vaginal seal member is formed at the in vitro end of the main assembly. The vaginal seal member includes a central axis coaxial to the first central axis. The vaginal seal member includes a contact wall and a non-contact wall opposite to the contact wall. The contact wall of the vaginal seal member is configured in such a way as to face and contact with at least a portion of a vulva of the patient. The vaginal seal member is configured to hermetically isolate the vaginal cavity of the patient when the contact wall of the vaginal seal member is positioned to be in contact with at least a portion of the vulva of the patient. Alternatively or in addition, the vaginal seal member is configured to ensure the main assembly is properly positioned in the vaginal cavity of the patient.

(12) The hysteroscopic system also includes a uterine assembly. The uterine assembly is formed as an elongated body having an in vivo end and an in vitro end. The uterine assembly is configured to be slidably housed in the main channel of the main assembly in such a way that the in vivo end of the uterine assembly is extendible outwardly away from the in vivo end of the main assembly (e.g., the uterine assembly can slide into and out of the main channel of the main assembly in both directions). The uterine assembly includes a first negative pressure port. The first negative pressure port is formed at a distal end of the in vivo end of the uterine assembly. The first negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the first negative pressure port is a state in which the first negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a second negative pressure. For example, such second negative pressure may be an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly. The non-haemostasis state of the first negative pressure port is a state in which the first negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the second negative pressure (e.g., the non-haemostasis state of the second negative pressure port may be a state in which the second negative pressure port does not apply any negative pressure). The uterine assembly also includes a uterine expandable member. The uterine expandable member is formed on (and/or attached to, formed with, etc.) the elongated body of the uterine assembly and adjacent to the first negative pressure port. The uterine expandable member is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the uterine expandable member is a state in which the uterine expandable member has a third overall volume (e.g., expanded outwardly away from the elongated body of the uterine assembly towards the wall of the uterine cavity). The non-haemostasis state of the uterine expandable member is a state in which the

uterine expandable member has a fourth overall volume (e.g., not expanded outwardly away from the elongated body of the uterine assembly). The third overall volume greater than the fourth overall volume. The uterine assembly also includes a second negative pressure port. The second negative pressure port is formed adjacent to the uterine expandable member in such a way that the uterine expandable member is positioned between the first and second negative pressure ports. The second negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the second negative pressure port is a state in which the second negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a third negative pressure. For example, such third negative pressure may be an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly. The non-haemostasis state of the second negative pressure port is a state in which the second negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the third negative pressure (e.g., the non-haemostasis state of the second negative pressure port may be a state in which the second negative pressure port does not apply any negative pressure).
(13) In another exemplary embodiment a hysteroscopic system for managing post-partum hemorrhaging is described. The hysteroscopic system includes a main assembly. The main assembly is formed as an elongated body having an in vivo end and an in vitro end. The in vivo end of the main assembly is configured to be inserted into and housed in a vaginal cavity of a patient. The in vitro end of the main assembly is configured to remain outside of the vaginal cavity of the patient. The main assembly includes a main channel. The main channel is formed through the elongated body of the main assembly between the in vivo and in vitro ends of the main assembly. The main channel is formed along a first central axis (and/or is formed in such a way as to include the first central axis formed through it). The main assembly also includes a first anchoring member. The first anchoring member is formed on the elongated body of the main assembly. The first anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the first anchoring member is a state in which the first anchoring member is expanded away from the elongated body of the main assembly. The main assembly also includes a second anchoring member. The second anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the second anchoring member is a state in which the second anchoring member is expanded away from the elongated body of the main assembly. The main assembly also includes a third anchoring member. The third anchoring member is formed between the first and second anchoring members. The third anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the third anchoring member is a state in which the third anchoring member applies at least a first negative pressure (e.g., suction).
(14) The hysteroscopic system also includes a uterine assembly. The uterine assembly is formed as an elongated body having an in vivo end and an in vitro end. The uterine assembly is configured to be slidably housed in the main channel of the main assembly (e.g., the uterine assembly can slide into and out of the main channel of the main assembly in both directions). The uterine assembly includes a first negative pressure port. The first negative pressure port is formed at or near a distal end of the in vivo end of the uterine assembly. The first negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the first negative pressure port is a state in which the first negative pressure port applies at least a second negative pressure. The uterine assembly also includes a uterine expandable member. The uterine expandable member is formed on the elongated body of the uterine assembly and adjacent to the first negative pressure port. The uterine expandable member is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the uterine expandable member is a state in which the uterine expandable member is expanded away from the elongated body of the uterine assembly. The uterine assembly also includes a second negative

pressure port. The second negative pressure port is formed adjacent to the uterine expandable member in such a way that the uterine expandable member is positioned between the first and second negative pressure ports. The second negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the second negative pressure port is a state in which the second negative pressure port applies at least a third negative pressure.

(15) The hysteroscopic system also includes a controller. The controller is configured to perform at least a transitioning of the first anchoring member between the anchoring state and the non-anchoring state. The controller is also configured to perform at least a transitioning of the second anchoring member between the anchoring state and the non-anchoring state. The controller is also configured to perform at least a transitioning of the third anchoring member between the anchoring state and the non-anchoring state. The controller is also configured to perform at least a transitioning of the first negative pressure port between the haemostasis state and the non-haemostasis state. The controller is also configured to perform at least a transitioning of the uterine expandable member between the haemostasis state and the non-haemostasis state. The controller is also configured to perform at least a transitioning of the second negative pressure port between the haemostasis state and the non-haemostasis state. The controller is also configured to apply the second and third negative pressures in such a way that a combination of the second and third negative pressures is a sufficient amount of negative pressure (and/or sufficiently directed in the right directions and/or placed in the right locations within the uterine cavity) so as to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly.

(16) In another exemplary embodiment a method of configuring a hysteroscopic system to manage post-partum hemorrhaging is described. The method includes providing a hysteroscopic system. The hysteroscopic system includes a main assembly. The main assembly is formed as an elongated body having an in vivo end and an in vitro end. The main assembly includes a main channel. The main channel is formed through the elongated body of the main assembly between the in vivo and in vitro ends of the main assembly. The main channel is formed along a first central axis (and/or is formed in such a way as to include the first central axis formed through it). The main assembly also includes a cervical seal member. The cervical seal member is formed at the in vivo end of the main assembly. The cervical seal member includes a central axis coaxial to the first central axis. The cervical seal member includes a contact wall and a non-contact wall opposite to the contact wall of the cervical seal member. The contact wall of the cervical seal member is configured in such a way as to face and contact with at least a portion of a cervix of the patient. The cervical seal member is configured to hermetically isolate a uterine cavity of the patient from the vaginal cavity of the patient when the contact wall of the cervical seal member is positioned to be in contact with at least a portion of the cervix of the patient. Alternatively or in addition, the cervical seal member is configured to ensure the main assembly does not enter into the uterine cavity of the patient. The main assembly also includes a first anchoring member. The first anchoring member is formed on (and/or attached to, formed with, etc.) the elongated body of the main assembly and adjacent to the cervical seal member. The first anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the first anchoring member is a state in which the first anchoring member has a first overall volume (e.g., expanded outwardly away from the elongated body of the main assembly towards the wall of the vaginal cavity). The non-anchoring state of the first anchoring member is a state in which the first anchoring member has a second overall volume (e.g., not expanded outwardly away from the elongated body of the main assembly). The first overall volume is greater than the second overall volume. The main assembly also includes a second anchoring member. The second anchoring member is formed on (and/or attached to) the elongated body of the main assembly. The second anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the

second anchoring member is a state in which the second anchoring member has a third overall volume (e.g., expanded outwardly away from the elongated body of the main assembly towards the wall of the vaginal cavity). The non-anchoring state of the second anchoring member is a state in which the second anchoring member has a fourth overall volume (e.g., not expanded outwardly away from the elongated body of the main assembly). The third overall volume is greater than the fourth overall volume. The first overall volume may (or may not) be the same volume as the third overall volume. The second overall volume may (or may not) be the same volume as the fourth overall volume. The main assembly also includes a third anchoring member. The third anchoring member is formed between the first and second anchoring members. The third anchoring member is configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the third anchoring member is a state in which the third anchoring member applies at least a first negative pressure (e.g., suction). The first negative pressure is an amount of negative pressure required to collapse at least a portion of the vaginal cavity (or vaginal wall forming the vaginal cavity) towards the elongated body of the main assembly. The non-anchoring state of the third anchoring member is a state in which the third anchoring member does not apply at least the first negative pressure (e.g., the non-anchoring state of the third anchoring member may be a state in which the third anchoring member does not apply any negative pressure). The hysteroscopic system also includes a uterine assembly. The uterine assembly is formed as an elongated body having an in vivo end and an in vitro end. The uterine assembly is configured to be slidably housed in the main channel of the main assembly in such a way that the in vivo end of the uterine assembly is extendible outwardly away from the in vivo end of the main assembly (e.g., the uterine assembly can slide into and out of the main channel of the main assembly in both directions). The uterine assembly includes a first negative pressure port. The first negative pressure port is formed at or near a distal end of the in vivo end of the uterine assembly. The first negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the first negative pressure port is a state in which the first negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a second negative pressure. For example, such second negative pressure may be an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly. The non-haemostasis state of the first negative pressure port is a state in which the first negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the second negative pressure (e.g., the non-haemostasis state of the first negative pressure port may be a state in which the first negative pressure port does not apply any negative pressure). The uterine assembly also includes a uterine expandable member. The uterine expandable member is formed on (and/or attached to, formed with, etc.) the elongated body of the uterine assembly and adjacent to the first negative pressure port. The uterine expandable member is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the uterine expandable member is a state in which the uterine expandable member has a third overall volume (e.g., expanded outwardly away from the elongated body of the uterine assembly towards the wall of the uterine cavity). The non-haemostasis state of the uterine expandable member is a state in which the uterine expandable member has a fourth overall volume (e.g., not expanded outwardly away from the elongated body of the uterine assembly). The third overall volume is greater than the fourth overall volume. The uterine assembly also includes a second negative pressure port. The second negative pressure port is formed adjacent to the uterine expandable member in such a way that the uterine expandable member is positioned between the first and second negative pressure ports. The second negative pressure port is configured to transition between a haemostasis state and a non-haemostasis state. The haemostasis state of the second negative pressure port is a state in which the second negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a third negative pressure. For example, such third negative pressure may be an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly. The non-anchoring state of the second negative pressure port is a state in which the second negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the third negative pressure (e.g., the non-haemostasis state of the second negative pressure port may be a state in which the second negative pressure port does not apply any negative pressure). The uterine assembly is configured to apply the second and third negative pressures in such a way that a combination of the second and third negative pressures is a sufficient amount of negative pressure (and/or sufficiently directed in the right directions and/or placed in the right locations within the uterine cavity) so as to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body of the uterine assembly.

(17) The method also includes configuring the uterine assembly and the main assembly in such a way that at least a portion of the uterine assembly is housed in the main channel of the main assembly. The method also includes introducing the hysteroscopic system trans-vaginally until the cervical seal member is brought into contact with at least a portion of a cervix. The contact of the cervical seal member and at least a portion of a cervix limits the air to pass in or out. The method also includes transitioning the main assembly to be in the anchoring state. The transitioning of the main assembly to be in the anchoring state causes at least a portion of the vaginal cavity to collapse towards the elongated body of the main assembly. The transitioning of the main assembly to be in the anchoring state includes transitioning the first anchoring member to be in the anchoring state. The transitioning of the main assembly to be in the anchoring state also includes transitioning the second anchoring member to be in the anchoring state. The transitioning of the main assembly to be in the anchoring state also includes transitioning the third anchoring member to be in the anchoring state. The method also includes transitioning the uterine assembly to be in the haemostasis state. The transitioning of the uterine assembly causes at least a portion of the uterine cavity to collapse towards the elongated body. The transitioning of the uterine assembly to be in the haemostasis state includes transitioning the uterine expandable member to be in the haemostasis state. The transitioning of the uterine assembly to be in the haemostasis state also includes transitioning the first negative pressure port to be in the haemostasis state. The transitioning of the uterine assembly to be in the haemostasis state also includes transitioning the second negative pressure port to be in the haemostasis state. The method also includes transitioning the uterine assembly from the haemostasis state to the non-haemostasis state. The method also includes removing the uterine assembly from the main channel of the main assembly. The transitioning of the uterine assembly from the haemostasis state to the non-haemostasis state includes transitioning the uterine expandable member to be in the non-haemostasis state. The transitioning of the uterine assembly from the haemostasis state to the non-haemostasis state also includes transitioning the first negative pressure port to be in the non-haemostasis state. The transitioning of the uterine assembly from the haemostasis state to the non-haemostasis state also includes transitioning the second negative pressure port to be in the non-haemostasis state.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) For a more complete understanding of the present disclosure, example embodiments, and their advantages, reference is now made to the following description taken in conjunction with the accompanying figures, in which like reference numbers indicate like features, and:

(2) FIG. **1**A is an illustration of a perspective view of an example embodiment of a hysteroscopic system (with main assembly in non-anchoring state and uterine assembly in non-haemostasis state) for managing post-partum hemorrhaging;

(3) FIG. **1**B is another illustration of a perspective view of an embodiment of a hysteroscopic system (with main assembly in non-anchoring state and uterine assembly in non-haemostasis state) for managing post-partum hemorrhaging;
(4) FIG. **1**C is another illustration of a perspective view of an embodiment of a hysteroscopic system (with main assembly in non-anchoring state and uterine assembly in non-haemostasis state) for managing post-partum hemorrhaging;
(5) FIG. **1**D is another is an illustration of a perspective view of an embodiment of a hysteroscopic system (with main assembly in non-anchoring state and uterine assembly in non-haemostasis state) for managing post-partum hemorrhaging;
(6) FIG. **1**E is another illustration of a perspective view of an example embodiment of a hysteroscopic system (with main assembly in anchoring state and uterine assembly in haemostasis state) for managing post-partum hemorrhaging;
(7) FIG. **1**F is another illustration of a perspective view of an example embodiment of a hysteroscopic system (with main assembly in anchoring state and uterine assembly in haemostasis state) for managing post-partum hemorrhaging;
(8) FIG. **1**G is another illustration of a perspective view of an example embodiment of a hysteroscopic system (with main assembly in anchoring state and uterine assembly in haemostasis state) for managing post-partum hemorrhaging;
(9) FIG. **1**H is another illustration of a perspective view of an example embodiment of a hysteroscopic system (with main assembly in anchoring state and uterine assembly in haemostasis state) for managing post-partum hemorrhaging;
(10) FIG. **2**A is an illustration of a side view of an example embodiment of a hysteroscopic system (with main assembly in non-anchoring state and uterine assembly in non-haemostasis state) for managing post-partum hemorrhaging;
(11) FIG. **2**B is another illustration of a side view of an example embodiment of a hysteroscopic system (with main assembly in anchoring state and uterine assembly in haemostasis state) for managing post-partum hemorrhaging;
(12) FIG. **2**C is another illustration of a side view of an example embodiment of a hysteroscopic system (with main assembly in non-anchoring state and uterine assembly in non-haemostasis state) for managing post-partum hemorrhaging;
(13) FIG. **2**D is another illustration of a side view of an example embodiment of a hysteroscopic system (with main assembly in anchoring state and uterine assembly in haemostasis state) for managing post-partum hemorrhaging;
(14) FIG. **3**A is an illustration of a side view of an example embodiment of a main assembly (in non-anchoring state) of a hysteroscopic system for managing post-partum hemorrhaging;
(15) FIG. **3**B is another illustration of a side view of an example embodiment of a main assembly (in non-anchoring state) of a hysteroscopic system for managing post-partum hemorrhaging;
(16) FIG. **3**C is another illustration of a side view of an example embodiment of a main assembly (anchoring state) of a hysteroscopic system for managing post-partum hemorrhaging;
(17) FIG. **3**D is another illustration of a side view of an example embodiment of a main assembly (non-anchoring state) of a hysteroscopic system for managing post-partum hemorrhaging;
(18) FIG. **4**A is an illustration of a main assembly (anchoring state) having an example embodiment of a cervical seal member with a shallow concave or dish-like shaped contact wall;
(19) FIG. **4**B is an illustration of a main assembly (anchoring state) having another example embodiment of a cervical seal member with a deeper concave or bowl-like shaped contact wall;
(20) FIG. **4**C is an illustration of a main assembly (anchoring state) having another example embodiment of a cervical seal member with an oval or convex shaped contact wall;
(21) FIG. **4**D is an illustration of a main assembly (anchoring state) having another example embodiment of a cervical seal member with a spherical or convex shaped contact wall;
(22) FIG. **4**E is an illustration of a side view of a main assembly;

(23) FIG. **4**F is an illustration of a front view of a main assembly having a main channel;
(24) FIG. **4**G is illustration of a side view of a main assembly;
(25) FIG. **4**H is illustration of a front view of a main assembly having one or more suction modules on the cervical seal member;
(26) FIG. **5**A is an illustration of a side view of an example embodiment of a uterine assembly (non-haemostasis state);
(27) FIG. **5**B is an illustration of a side view of an example embodiment of a uterine assembly (hemostasis state);
(28) FIG. **5**C is an illustration of a side view of an example embodiment of a uterine assembly having a bendable member;
(29) FIG. **5**D is an illustration of a side view of an example embodiment of a uterine assembly having a second extendible member;
(30) FIG. **6**A is an illustration of an example embodiment of a hysteroscopic system being inserted into a vaginal cavity of a patient;
(31) FIG. **6**B is an illustration of an example embodiment of a hysteroscopic system having a main assembly (non-anchoring state) housed in a vaginal cavity of a patient and a uterine assembly (non-hemostasis state) housed in a uterine cavity of the patient;
(32) FIG. **6**C is an illustration of an example embodiment of a hysteroscopic system having a main assembly (first and second anchoring members in the anchoring state) housed in a vaginal cavity of a patient and a uterine assembly (non-hemostasis state) housed in a uterine cavity of the patient;
(33) FIG. **6**D is an illustration of an example embodiment of a hysteroscopic system having a main assembly (first, second, and third anchoring members in the anchoring state) anchored to a vaginal cavity of a patient and a uterine assembly (non-hemostasis state) housed in a uterine cavity of the patient;
(34) FIG. **6**E is an illustration of an example embodiment of a hysteroscopic system having a main assembly (first, second, and third anchoring members in the anchoring state) anchored to a vaginal cavity of a patient and a uterine assembly (first negative pressure port, uterine expandable member, and second negative pressure port transitioned to the hemostasis state) housed in a uterine cavity of the patient;
(35) FIG. **6**F is an illustration of an example embodiment of a hysteroscopic system having a main assembly (first, second, and third anchoring members in the anchoring state) anchored to a vaginal cavity of a patient and a uterine assembly (first negative pressure port, uterine expandable member, and second negative pressure port in the hemostasis state) anchored to a uterine cavity of the patient;
(36) FIG. **6**G is an illustration of an example embodiment of a hysteroscopic system having a main assembly (first, second, and third anchoring members in the anchoring state) anchored to a vaginal cavity of a patient and a uterine assembly (first negative pressure port, uterine expandable member, and second negative pressure port in the hemostasis state) anchored to a uterine cavity of the patient, wherein the anchored uterine assembly is moved relative to the anchored main assembly;
(37) FIG. **6**H is an illustration of an example embodiment of a hysteroscopic system having a main assembly (first, second, and third anchoring members in the anchoring state) anchored to a vaginal cavity of a patient and a uterine assembly (non-hemostasis state) being removed from uterine and vaginal cavities of the patient;
(38) FIG. **6**I is an illustration of an example embodiment of a hysteroscopic system having a main assembly (first, second, and third anchoring members in the anchoring state) anchored to a vaginal cavity of a patient and a manipulator assembly inserted into a uterine cavity of the patient to perform a surgical action in the uterine cavity of the patient;
(39) FIG. **7**A is an illustration of a perspective view of an example embodiment of a locking mechanism;
(40) FIG. **7**B is an illustration of a front view of an example embodiment of a locking mechanism

configurable or configured to transition between locked and unlocked states based on a rotation direction of the locking mechanism;
(41) FIG. **7**C is another illustration of front view of an example embodiment of locking mechanism in the locked state;
(42) FIG. **8**A is an illustration of a perspective view of an example embodiment of a locking mechanism in a locked state;
(43) FIG. **8**B is an illustration of a perspective view of an example embodiment of a locking mechanism in an unlocked state;
(44) FIG. **8**C is an illustration of a front view of an example embodiment of a locking mechanism in a locked state; and
(45) FIG. **8**D is an illustration of a front view of an example embodiment of a locking mechanism in an unlocked state.
(46) Although similar reference numbers may be used to refer to similar elements in the figures for convenience, it can be appreciated that each of the various example embodiments may be considered to be distinct variations.
(47) Example embodiments will now be described with reference to the accompanying figures, which form a part of the present disclosure and which illustrate example embodiments which may be practiced. As used in the present disclosure and the appended claims, the terms “embodiment”, “example embodiment”, “exemplary embodiment”, and “present embodiment” do not necessarily refer to a single embodiment, although they may, and various example embodiments may be readily combined and/or interchanged without departing from the scope or spirit of example embodiments. Furthermore, the terminology as used in the present disclosure and the appended claims is for the purpose of describing example embodiments only and is not intended to be limitations. In this respect, as used in the present disclosure and the appended claims, the term “in” may include “in” and “on”, and the terms “a”, “an”, and “the” may include singular and plural references. Furthermore, as used in the present disclosure and the appended claims, the term “by” may also mean “from,” depending on the context. Furthermore, as used in the present disclosure and the appended claims, the term “if” may also mean “when” or “upon”, depending on the context. Furthermore, as used in the present disclosure and the appended claims, the words “and/or” may refer to and encompass any and all possible combinations of one or more of the associated listed items.
DETAILED DESCRIPTION
(48) Present example embodiments relate generally to and/or include systems, subsystems, assemblies, processors, and devices for addressing conventional problems, including those described above and in the present disclosure, and more specifically, example embodiments relate to systems, subsystems, assemblies, processors, controllers, and devices for managing postpartum hemorrhaging.
(49) It is to be understood in the present disclosure that example embodiments may also be used to manage, control, reduce, treat, stop, or the like (referred to herein as “manage”) other clinical conditions including, but not limited to, obstetric hemorrhaging, post-abortion hemorrhaging, vaginal bleeding, uterine atony, cervical laceration, uterine perforation, cervical pregnancy, etc.
(50) Example embodiments will now be described below with reference to the accompanying figures, which form a part of the present disclosure.
(51) Example Embodiments of a Hysteroscopic System for Managing Postpartum Hemorrhaging (e.g., Hysteroscopic System **100**).
(52) FIG. **1**A, FIG. **1**B, FIG. **1**C, FIG. **1**D, FIG. **1**E, FIG. **1**F, FIG. **1**G, FIG. **1**H, FIG. **2**A, FIG. **2**B, FIG. **2**C, and FIG. **2**D illustrate example embodiments of a hysteroscopic system (e.g. hysteroscopic system **100**). The hysteroscopic system **100** may be configurable or configured to manage postpartum hemorrhaging (or “PPH”). Alternatively or in addition, the hysteroscopic system **100** may be configurable or configured to manage other clinical conditions including, but

not limited to, obstetric hemorrhaging, post-abortion hemorrhaging, vaginal bleeding, uterine atony, cervical laceration, uterine perforation, cervical pregnancy. Alternatively or in addition, the hysteroscopic system **100** may be configurable or configured to manage other clinical conditions using suction and/or anchorage mechanism.

(53) To perform the actions, functions, processes, and/or methods described above and in the present disclosure, example embodiments of the hysteroscopic system **100** include one or more elements.

(54) For example, the hysteroscopic system **100** includes a main assembly (e.g., main assembly **200**). The main assembly **200** may be formed as and/or including an elongated body **201** having an in vivo end **200***a* and an in vitro end **200***b*. The main assembly **200** may include a main channel **202** formed through the elongated body **201** of the main assembly **200**. The main assembly **200** may also include one or more cervical seal members **210** formed at the in vivo end **200***a* of the main assembly **200**. The cervical seal member **210** may be configurable or configured to, among other things, ensure that the main assembly **200** is properly positioned in a vaginal cavity of the patient (e.g., by ensuring that the cervical seal member **210** abuts, is adjacent to, seals, encloses, and/or is in contact with at least a portion of a cervix (and/or surround areas of the cervix) of the patient) and/or ensure that the uterine cavity and the vaginal cavity of the patient are isolated from one another (e.g., by providing a separation, seal, and/or hermetic seal for the uterine cavity). The main assembly **200** may also include one or more first anchoring members **220***a*. The main assembly **200** may also include one or more second anchoring members **220***b*. The main assembly **200** may also include one or more third anchoring members **230**. The main assembly **200** may also include one or more locking members or mechanisms **240**. The locking member **240** may be configurable or configured to join, lock, restrict, or the like, movement of the main assembly **200** relative to another element of the hysteroscopic system **100** (e.g., relative to the uterine assembly **300** when the uterine assembly **300** is inserted through the main channel **202**; relative to the manipulator assembly **400** when the manipulator assembly **400** is inserted through the main channel **202**; etc.). The main assembly **200** may also include one or more vaginal seal members **250**. The vaginal seal member **250** may be configurable or configured to, among other things, ensure that the main assembly **200** is properly positioned in a vaginal cavity of the patient (e.g., by ensuring that the vaginal seal member **250** abuts, is adjacent to, seals, encloses, and/or is in contact with at least a portion of a vulva (and/or surrounding areas of the vulva) of the patient) and/or ensure that the vaginal cavity of the patient is separated and/or hermetically sealed (e.g., separated from outside of the patient).

(55) As will be further described in the present disclosure, when in operation, the in vivo end **200***a* of the main assembly **200** is configurable or configured to be inserted into (e.g., through a patient's vagina) and housed in a vaginal cavity of the patient. The in vitro end **200***b* of the main assembly **200** is configurable or configured to remain outside of the vaginal cavity of the patient. As will be further described in the present disclosure, the main assembly **200** is configurable or configured to be anchored and/or secured in the vaginal cavity of the patient (and/or anchored and/or secured relative to the vaginal cavity of the patient and/or vaginal walls forming the vaginal cavity of the patient; also referred to herein as an “anchoring state” or “anchored state”) through one or more other elements of the hysteroscopic system **100** (e.g., via transitioning of the first anchoring member **220***a* to an anchoring state; transitioning of the second anchoring member **220***b* to an anchoring state; and/or transitioning of the third anchoring member **230** to an anchoring state). The main assembly **200** is also configurable or configured to be non-anchored and/or non-secured (or not anchored or not secured) in the vaginal cavity of the patient (and/or non-anchored and/or non-secured relative to the vaginal cavity of the patient and/or vaginal walls forming the vaginal cavity of the patient; also referred to herein as an “non-anchoring state” or “non-anchored state”) through one or more other elements of the hysteroscopic system **100** (e.g., via transitioning of the first anchoring member **220***a* to a non-anchoring state; transitioning of the second anchoring member

220*b* to a non-anchoring state; and/or transitioning of the third anchoring member **230** to a non-anchoring state).

(56) As used in the present disclosure, a reference to “anchor”, “anchoring”, “anchored”, “anchorable”, “secure”, “securing”, “secured”, “securable”, or the like, of one or more elements of the hysteroscopic system **100** (e.g., the main assembly **200**, first anchoring member **220***a*, second anchoring member **220***b*, third anchoring member **230**, uterine assembly **300**, first negative pressure port **310**, uterine expandable member **320**, and/or second negative pressure port **330**) to one or more cavities and/or walls forming a cavity of a patient (e.g., vaginal cavity, vaginal wall forming a vaginal cavity of the patient, uterine cavity, and/or uterine wall forming a uterine cavity of the patient) includes setting, changing, or transitioning a state of one or more elements of the hysteroscopic system **100** (e.g., transitioning from a non-anchoring state to an anchoring state; transitioning from a non-hemostasis state to a hemostasis state; transitioning from a non-securing state to a securing state; etc.); actuating, adjusting, expanding, and/or protruding one or more elements of the hysteroscopic system **100** (e.g., for expandable members, such as the first anchoring member **220***a*, second anchoring member **220***b*, uterine expandable member **320**, etc.); setting, applying, or increasing negative pressure or suction from one or more elements of the hysteroscopic system **100** (e.g., for negative pressure ports, such as the third anchoring member **230**, first negative pressure port **310**, second negative pressure port **330**, etc.); or the like, in such a way that such one or more elements of the hysteroscopic system **100** (e.g., the main assembly **200**, uterine assembly **300**, etc.) are not readily or easily movable, slidable, or the like, relative to the cavity and/or wall forming the cavity of the patient (e.g., not readily or easily movable, slidable, or the like, as compared to when the element of the hysteroscopic system **100** is not anchored; and such anchoring may be able to withstand a force of between 10N to 50N applied to such one or more elements of the hysteroscopic system **100**).

(57) The hysteroscopic system **100** also includes a uterine assembly (e.g., uterine assembly **300**). The uterine assembly **300** may be formed as an elongated body **301** having an in vivo end **300***a* and an in vitro end **300***b*. The uterine assembly **300** may include one or more first negative pressure ports **310**. The uterine assembly **300** may also include one or more uterine expandable members **320**. The uterine assembly **300** may also include one or more second negative pressure ports **330**. The uterine assembly **300** may also include one or more locking members or mechanisms (not shown). The locking member for the uterine assembly **300** may be configurable or configured to join, lock, restrict, or the like, movement of the uterine assembly **300** relative to the main assembly **200** (and may do so either in cooperation with the locking member **240** of the main assembly **200** or without use of the locking member **240** of the main assembly **200**). The uterine assembly **300** is configurable or configured to cooperate with the main assembly **200** in performing a surgical action, including managing of PPH. As will be further described in the present disclosure, when in operation, the uterine assembly **300** is configurable or configured to be inserted into a main channel **202** of the main assembly **200** (from the in vitro end **200***b* of the main assembly **200**) in such a way that the in vivo end **300***a* of the uterine assembly **300** is positioned outwardly away from the in vivo end **200***a* of the main assembly **200**. In this regard, when the in vivo end **300***a* of the uterine assembly **300** is positioned outwardly away from the in vivo end **200***a* of the main assembly **200** (i.e., not housed in the main channel **202** of the main assembly **200**, but housed in a uterine cavity of a patient (e.g., see at least FIGS. **6**B-G)), at least a portion of the elongated body **301** of the uterine assembly **300** remains housed in the main channel **202** of the main assembly **200** (e.g., see at least FIGS. **6**B-G). Furthermore, at least a portion of the in vitro end **300***b* of the uterine assembly **300** is not housed in the main channel **202** of the main assembly **200**. As will be further described in the present disclosure, the uterine assembly **300** is configurable or configured to be anchored and/or secured in the uterine cavity of the patient (and/or anchored and/or secured relative to the uterine cavity of the patient and/or uterine walls forming the uterine cavity of the patient; also referred to herein as a “hemostasis state” or “anchored state”) through one or more other

elements of the hysteroscopic system **100** (e.g., via transitioning of the first negative pressure port **310** to a hemostasis state or anchored state; transitioning of the uterine expandable member to a hemostasis state or anchored state; and/or transitioning of the second negative pressure port to a hemostasis state or anchored state). The uterine assembly **300** is also configurable or configured to be non-anchored and/or non-secured (or not anchored or not secured) in the uterine cavity of the patient (and/or non-anchored and/or non-secured relative to the uterine cavity of the patient and/or uterine walls forming the uterine cavity of the patient; referred to herein as the "non-hemostasis state") through one or more other elements of the hysteroscopic system **100** (e.g., via transitioning of the first negative pressure port **310** to a non-hemostasis state or non-anchoring state; transitioning of the uterine expandable member to a non-hemostasis state or non-anchoring state; and/or transitioning of the second negative pressure port to a non-hemostasis state or non-anchoring state).

(58) The hysteroscopic system **100** may also include a manipulator assembly (e.g., manipulator assembly **400**). The manipulator assembly **400** may be formed as an elongated body **401** having an in vivo end **400***a* and an in vitro end **400***b*. The manipulator assembly **400** may include one or more end effectors **410**. The manipulator assembly **400** may also include one or more expandable members (not shown) similar to or the same as the uterine expandable member **320** of the uterine assembly **300**. The manipulator assembly **400** may also include one or more negative pressure ports (not shown) similar to or the same as the first negative pressure port **310** and/or second negative pressure port **330** of the uterine assembly **300**. The manipulator assembly **400** may also include one or more locking members or mechanisms (not shown). The locking member for the manipulator assembly **400** may be configurable or configured to join, lock, restrict, or the like, movement of the manipulator assembly **400** relative to the main assembly **200** (and may do so either in cooperation with the locking member **240** of the main assembly **200** or without use of the locking member **240** of the main assembly **200**). The manipulator assembly **400** is configurable or configured to cooperate with the main assembly **200** in performing a surgical action, including managing of PPH. As will be further described in the present disclosure, when in operation, the manipulator assembly **400** is configurable or configured to be inserted into a main channel **202** of the main assembly **200** (from the in vitro end **200***b* of the main assembly **200**) in such a way that the in vivo end **400***a* of the manipulator assembly **400** is positioned outwardly away from the in vivo end **200***a* of the main assembly **200** so as to allow an end effector **410** of the manipulator assembly **400** to perform a surgical action in the uterine cavity of the patient. In this regard, when the in vivo end **400***a* of the manipulator assembly **400** is positioned outwardly away from the in vivo end **200***a* of the main assembly **200** (i.e., not housed in the main channel **202** of the main assembly **200**, but housed in a uterine cavity of a patient (e.g., see at least FIG. **6**I)), at least a portion of the elongated body **401** of the manipulator assembly **400** remains housed in the main channel **202** of the main assembly **200** (e.g., see at least FIG. **6**I). Furthermore, at least a portion of the in vitro end **400***b* of the manipulator assembly **400** is not housed in the main channel **202** of the main assembly **200**.

(59) In an example embodiment, the uterine assembly **300** is a separate assembly from the main assembly **200**. When in operation, the main assembly **200** is insertable into and anchorable in the vaginal cavity of the patient (and/or anchorable relative to the vaginal cavity of the patient and/or vaginal walls forming the vaginal cavity of the patient) (e.g., via the first anchoring member **220***a*, second anchoring member **220***b*, and/or third anchoring member **230**). As will be further described in the present disclosure, a cervical seal member **210** is configurable or configured to, among other things, assist in properly positioning the main assembly **200** in the vaginal cavity of the patient.

(60) Once the main assembly **200** is properly positioned and anchored in the vaginal cavity of the patient, the uterine assembly **300** is then insertable (or slidable) through the main channel **202** (from the in vitro end **200***b* of the main assembly **200**) until the in vivo end **300***a* of the uterine assembly **300** passes through the main channel **220** and into the uterine cavity of the patient.

(61) Once at least a portion of the in vivo end **300***a* of the uterine assembly **300** is provided or

housed in the uterine cavity of the patient (e.g., the at least one portion of the in vivo end **300***a* of the uterine assembly **300** that includes the first negative pressure port **310**, the uterine expandable member **320**, and the second negative pressure port **330**), the uterine assembly **300** is transitionable from a non-hemostasis state (or non-anchoring state) to a hemostasis state (or anchoring state), which is a state in which the hysteroscopic system **100** can commence managing PPH of the patient. More specifically, the hemostasis state (or anchoring state) of the uterine assembly **300** may be a state in which the first negative pressure port **310** has transitioned to a hemostasis state (or anchoring state) (e.g., the hemostasis state (or anchoring state) of the first negative pressure port **310** may be a state in which the first negative pressure port **310** applies a negative pressure having a magnitude greater than or equal to a threshold value (such threshold value being a sufficient amount of negative pressure to encourage, bring in, urge, and/or collapse at least a portion of the uterine cavity and/or uterine wall forming the uterine cavity towards the elongated body **301** of the uterine assembly **300**)); the uterine expandable member **320** has transitioned to a hemostasis state (or anchoring state) (e.g., the hemostasis state (or anchoring state) of the uterine expandable member **320** may be a state in which the uterine expandable member **320** is expanded outwardly away from the elongated body **301** of the uterine assembly **300**); and/or the second negative pressure port **330** has transitioned to a hemostasis state (or anchoring state) (e.g., the hemostasis state (or anchoring state) of the second negative pressure port **330** may be a state in which the second negative pressure port **330** applies a negative pressure having a magnitude greater than or equal to a threshold value (such threshold value being a sufficient amount of negative pressure to encourage, bring in, urge, and/or collapse at least a portion of the uterine cavity and/or uterine wall forming the uterine cavity towards the elongated body **301** of the uterine assembly **300**)).
(62) In an example embodiment, the hemostasis state (or anchoring state) of the uterine assembly **300** may be a state in which a combination of the negative pressures applied by the first negative pressure port **310** and second negative pressure port **320** is a sufficient amount of negative pressure required to encourage, bring in, urge, and/or collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body **301** of the uterine assembly **300**.
(63) Once the uterine assembly **300** is transitioned to the hemostasis state (or anchoring state) (e.g., when the first negative pressure port **310**, uterine expandable member **320**, and second negative pressure port **330** are transitioned to the hemostasis state (or anchoring state)), the first negative pressure port **310** and the second negative pressure port **330** are configurable or configured in such a way that the collective negative pressure from the first negative pressure port **310** and the second negative pressure port **330** is sufficient to cause the uterine cavity and/or the uterine walls forming the uterine cavity to collapse, cave-in, envelope, shrink, or the like, inwards towards the uterine expandable member **320** (which is also transitioned to the hemostasis state (or anchoring state)). A locking mechanism **240** (and/or locking mechanism (not shown) of the uterine assembly **300**) may be further configurable or configured to secure or lock the main assembly **200** (and/or movement and/or position of the main assembly **200**) relative to the uterine assembly **300** so as to enable the uterine assembly **300** to be held in the homeostasis state (or anchoring state) for a prolonged period of time to manage (e.g., stop) the PPH for the patient (e.g., for between 5 minutes to over 24 hours).
(64) As used in the present disclosure, when applicable, a reference to a hysteroscopic system **100** (and/or one or more of its elements), surgeon console (not shown) (and/or one or more of its elements), and/or controller (not shown) (and/or one or more of its elements) may also refer to, apply to, and/or include one or more computing devices, processors, servers, systems, cloud-based computing, or the like, and/or functionality of one or more processors, computing devices, servers, systems, cloud-based computing, or the like. The hysteroscopic system **100** (and/or one or more of its elements), surgeon console (and/or one or more of its elements), and/or controller (and/or one or more of its elements) may be or have any processor, server, system, device, computing device,

controller, microprocessor, microcontroller, microchip, semiconductor device, or the like, configurable or configured to perform, among other things, positioning determination and/or control of one or more elements of the hysteroscopic system **100**; image capturing (including video image and still image capturing); image processing (including automated image processing of video images and still images); and/or any one or more other actions, functions, methods, and/or processes described above and in the present disclosure. Alternatively or in addition, the hysteroscopic system **100** (and/or one or more of its elements), surgeon console (and/or one or more of its elements), and/or controller (and/or one or more of its elements) may include and/or be a part of a virtual machine, processor, computer, node, instance, host, or machine, including those in a networked computing environment.
(65) These and other elements of the hysteroscopic system **100** will now be further described with reference to the accompanying figures.
(66) The Main Assembly (e.g., Main Assembly **200**).
(67) As illustrated in at least in FIG. **3**A, FIG. **3**B, FIG. **3**C, and FIG. **3**D, the hysteroscopic system **100** includes a main assembly (e.g., main assembly **200**). The main assembly **200** is configurable or configured to be inserted into and anchored in a vaginal cavity of a patient (and/or anchored relative to the vaginal cavity of the patient and/or vaginal walls forming the vaginal cavity of the patient).
(68) To perform the actions, functions, processes, and/or methods described above and in the present disclosure, example embodiments of the main assembly **200** include one or more elements.
(69) For example, the main assembly **200** includes and/or is formed as an elongated body **201** having an in vivo end **200***a* and an in vitro end **200***b* opposite to the in vivo end **200***a*. The elongated body **201** of the main assembly **200** may be formed in one or more of a variety of shapes, dimensions, configurations, etc. For example, the elongated body **201** of the main assembly **200** may be formed in the shape of a cylindrical tube, or the like, and/or with a circular, elliptical, oval, or other cross sectional shape.
(70) The main assembly **200** also includes a main channel **202**. The main channel **202** may be any channel, means, opening, passageway, pathway, tunnel, or the like, formed through the elongated body **201** of the main assembly **200** between the in vivo end **200***a* of the main assembly **200** and the in vitro end **200***b* of the main assembly **200**. The main channel **202** may be formed along a first central axis of the elongated body **201** of the main assembly **200**. The main assembly **200** also includes one or more cervical seal members **210**. The cervical seal member **210** may be configurable or configured to properly position the main assembly **200** in a vaginal cavity of a patient (e.g., by positioning the main assembly **200** in such a way that the cervical seal member **210** abuts, is adjacent to, seals, encloses, and/or is in contact with at least a portion of a cervix (and/or surround areas of the cervix) of the patient). Alternatively or in addition, the cervical seal member **210** may be configurable or configured to isolate the uterine cavity of the patient from the vaginal cavity of the patient (e.g., by providing a separation, seal, and/or hermetic seal for the uterine cavity). The main assembly **200** also includes one or more first anchoring members **220***a*. The first anchoring member **220***a* is configurable or configured to selectively expand (when in an anchoring state) and contract (when in a non-anchoring state). The main assembly **200** also includes one or more second anchoring members **220***b*. Similar to the first anchoring member **220***a*, the second anchoring member **220***b* is configurable or configured to selectively expand (when in an anchoring state) and contract (when in a non-anchoring state). The main assembly **200** also includes one or more third anchoring members **230**. The third anchoring member **230** is configurable or configured to provide a negative pressure (or suction). The main assembly **200** may also include one or more vaginal seal members **250** (e.g., as illustrated in at least FIGS. **2**C, **2**D, and **3**D). The main assembly **200** may also include one or more locking members or mechanisms **240** (e.g., as illustrated in at least FIG. **1**H). The main assembly **200** may also include one or more first extendible members **260** (e.g., as illustrated in at least FIG. **3**B).

(71) These and other elements of the main assembly **200** will now be further described with reference to the accompanying figures.
(72) The Elongated Body of the Main Assembly (e.g., Elongated Body **201**).
(73) As illustrated in at least FIGS. **1**A-**1**H and **2**A-**2**D, the main assembly **200** includes an elongated body of the main assembly (e.g., elongated body **201**). The elongated body **201** of the main assembly **200** may be configurable or configured to function as a core structure to provide structural rigidity for the main assembly **200**, and to position one or more elements of the hysteroscopic system **100** to perform one or more actions. The elongated body **201** of the main assembly **200** may be configurable or configured to support each element of the main assembly **200** (e.g. the cervical seal member **210**, the first anchoring member **220***a*, the second anchoring member **220***b*, the third anchoring member **230**, the locking member **240**, the vaginal seal member **250**, the first extendible member **260**, etc.). The elongated body **201** of the main assembly **200** is formed having sufficient rigidity so as to not bend, collapse, or deform when in operation, including not bend, collapse, or deform when: the cervical seal member **210** provides a separation, seal, and/or hermetic seal; the first anchoring member **220***a* is in the anchoring state (and/or is transitioned to and/or from the anchoring state); the second anchoring member **220***b* is in the anchoring state (and/or is transitioned to and/or from the anchoring state); the third anchoring member **230** is in the anchoring state (and/or is transitioned to and/or from the anchoring state); and/or when the vaginal seal member **250** provides a separation, seal, and/or hermetic seal.
(74) The elongated body **201** of main assembly **220** may be formed as a tubular shaped member, cylindrical shaped member, hollow member, square tubular shaped member, etc.
(75) The Cervical Seal Member (e.g., Cervical Seal Member **210**).
(76) As illustrated in at least FIGS. **1**A-**1**H, **2**A-**2**D, and **3**A-**3**D, and FIG. **4**A, FIG. **4**B, FIG. **4**C, FIG. **4**D, FIG. **4**E, FIG. **4**F, FIG. **4**G, and FIG. **4**H, the main assembly **200** includes one or more cervical seal members (e.g., cervical seal member **210**). The cervical seal member **210** may be configurable or configured to separate, isolate, seal, and/or hermetically seal a uterine cavity of a patient from a vaginal cavity of the patient. Alternatively or in addition, the cervical seal member **210** may be configurable or configured to ensure the main assembly **200** does not enter into the uterine cavity of the patient. Alternatively or in addition, the cervical seal member **210** may be configurable or configured to properly position the main assembly **200** within the vaginal cavity of the patient.
(77) The cervical seal member **210** may be formed at the in vivo end **200***a* of the main assembly **200**. In example embodiments, the cervical seal member **210** includes a central axis that is coaxial to the central axis of the main assembly **200** (and/or central axis of the main channel **202**). The cervical seal member **210** may be introduced trans-vaginally to abut, contact, seal, and/or hermetically seal a cervix (and/or surrounding areas of the cervix) of a patient. The cervical seal member **210** may be formed along a plane that is orthogonal or substantially orthogonal to the main channel **202** (and/or a central axis formed through the main channel **202**). The cervical seal member **210** may include an outermost edge portion (or perimeter portion or rim portion) formed between the contact wall **212** and non-contact wall **214** of the cervical seal member **210**. Such outermost edge portion of the cervical seal member **210** may be formed along a plane that is orthogonal or substantially orthogonal to the main channel **202** (and/or a central axis formed through the main channel **202**).
(78) To perform the actions, functions, processes, and/or methods described above and in the present disclosure, example embodiments of the cervical seal member **210** includes one or more elements and/or characteristics. For example, the cervical seal member **210** includes one or more contact walls **212**. In some embodiments, the cervical seal member **210** may also include one or more non-contact walls **214**. In some embodiments, the cervical seal member **210** may also include one or more sealable members **216** (e.g., a gel sheet, plate, or the like, capable of allowing one or more elements of the hysteroscopic system **100** (e.g., the uterine assembly **300**) to pass through,

while also providing a seal around the portion of the element that is passed through). In some embodiments, the cervical seal member **210** may also include one or more suction module **218** (e.g., negative pressure openings or ports that provide negative pressure or suction).

(79) The Contact Wall of the Cervical Seal Member (e.g., Contact Wall **212**).

(80) As illustrated in at least FIGS. **1**A-H and **4**A-H, the cervical seal member **210** includes one or more contact walls **212**. The contact wall **212** of the cervical seal member **210** may be configurable or configured to contact with at least a portion of the cervix of the patient. When contacting with at least a portion of the cervix of the patient, the contact wall **212** may separate, isolate, and/or hermetically seal the uterine cavity of the patient from the vaginal cavity of the patient (e.g., in cooperation with the sealable member **216** and/or one or more other elements of the hysteroscopic system **100**, such as the uterine assembly **300**). The contact wall **212** of the cervical seal member **210** may include a circular contact portion (and/or any other geometrical shape). The circular contact portion of the contact wall **212** may be formed along a plane that is substantially orthogonal to a central axis formed through the main channel **202** (also referred to herein as the first central axis). The contact wall **212** of the cervical seal member **210** may include at least a portion that is formed in a concave shape. Alternatively or in addition, the contact wall **212** of the cervical seal member **210** may include at least a portion that is formed in a convex shape, curve shape, rounded shape, etc. In some embodiments, the contact wall **212** of the cervical seal member **210** may be formed in a concave or dish-like shape (e.g., as illustrated in FIG. **4**A). In some embodiments, the contact wall **212** of the cervical seal member **210***a* may be formed in a deeper concave or bowl-like shape (e.g., as illustrated in FIG. **4**B). In some embodiments, the contact wall **212** of the cervical seal member **210***b* may be formed in a convex or oval-like shape (e.g., as illustrated in FIG. **4**C). In some embodiments, the contact wall **212** of the cervical seal member **210***c* may be formed in a convex or sphere-like shape (e.g., as illustrated in FIG. **4**D). In some embodiments, the contact wall **212** of the cervical seal member **210** may be formed in a ball-like shape (not shown) and/or other geometrical shapes (not shown) that can be closely contacted with at least a portion of the cervix of a patient. In some example embodiments, the contact wall **212** of the cervical seal member **210** is in a dish-like shape having a circular contact portion (e.g., as illustrated in FIGS. **4**E and **4**G). In some example embodiments, the circular contact portion of the contact wall **212** of the cervical seal member **210** may be formed along a plane that is orthogonal or substantially orthogonal to the main channel **202** (and/or a central axis formed through the main channel **202**) (e.g., as illustrated in FIG. **4**F). In some example embodiments, the circular contact portion of the contact wall **212** of the cervical seal member **210** may have one or more suction modules **218** on the cervical seal member **210** (e.g., as illustrated in FIG. **4**H). The contact wall **212** and/or non-contact wall **214** of the cervical seal member **210** may include one or more flexible portions (e.g., the outermost edge portion formed between the contact wall **212** and non-contact wall **214**). The at least one flexible portion of the contact wall **212** and/or non-contact wall **214** of the cervical seal member **210** is configured to dynamically adapt to a surface topology of the cervix of the patient (and/or provide a seal and/or hermetic seal).

(81) The Non-Contact Wall of the Cervical Seal Member (e.g., Non-Contact Wall **214**).

(82) In an example embodiment, the cervical seal member **210** may also include a non-contact wall **214**. The non-contact wall **214** of the cervical seal member **210** may be opposite to the contact wall **212** of the cervical seal member **210**. The non-contact wall **214** of the cervical seal member **210** may be adjacently positioned next to the first anchoring member **220***a*. The non-contact wall **214** of the cervical seal member **210** may include one or more portions formed in a concave shape. Alternatively or in addition, the non-contact wall **214** of the cervical seal member **210** may include at least a portion that is formed in a convex shape, curve shape, rounded shape, etc. In some example embodiments, the non-contact wall **214** of the cervical seal member **210** includes at least a portion that is formed in a convex shape. The non-contact wall **214** of the cervical seal member **210** may be formed along a plane that is orthogonal or substantially orthogonal to the main channel **202**

(and/or a central axis formed through the main channel **202**).

(83) The Sealable Member of the Cervical Seal Member (e.g., Sealable Member **216**).

(84) As illustrated in at least FIGS. **1**A-H, the cervical seal member **210** may also include one or more sealable members **216**. The sealable member **216** may be configurable or configured to dynamically adjust or transition between an opening state (e.g., a state in which one or more elements of the hysteroscopic system **100** are provided through the sealable member **216**) and a closed state (e.g., a state in which no elements of the hysteroscopic system **100** are provided through the sealable member **216**).

(85) For example, when a uterine assembly **300** is provided through the sealable member **216** (opening state), the sealable member **216** is configurable or configured to provide a hermetic seal around the uterine assembly **300** (or the portion of the uterine assembly **300** provided through the sealable member **216**). The sealable member **216** may be configurable or configured to allow the uterine assembly **300** to slide or pass through in both directions (opening state). As another example, when a manipulator assembly **400** is provided through the sealable member **216** (opening state), the sealable member **216** is configurable or configured to provide a hermetic seal around the manipulator assembly **400** (or the portion of the manipulator assembly **400** provided through the sealable member **216**). The sealable member **216** may be configurable or configured to allow the manipulator assembly **400** to slide or pass through in both directions (opening state). The sealable member **216** may be configurable or configured to hermetically seal itself when the manipulator assembly **400** is not provided through the sealable member **216** (closed state). The sealable member **216** may be formed coaxial to the cervical seal member **210** (and/or a central axis formed through the cervical seal member **210**). Alternatively or in addition, the sealable member **216** may be formed coaxial to the main channel **202** (and/or a central axis formed through the main channel **202**).

(86) The Suction Module of the Cervical Seal Member (e.g., Suction Module **218**).

(87) As illustrated in at least FIGS. **1**C and **4**H, the cervical seal member **210** may also include one or more suction modules **218**. The one or more suction modules **218** may be formed on the contact wall **212** of the cervical seal member **210**. The one or more suction modules **218** may be configurable or configured to provide a negative pressure. When in operation, such negative pressure may enable or facilitate a seal and/or hermetic seal between the cervical seal member **210** and a cervix of the patient (and/or uterine wall and/or vaginal wall of the patient). The one or more suction modules **218** may also enable or facilitate a collapse or compression at least a portion of the uterine wall, cervix, and/or vaginal wall of the patient.

(88) In an example embodiment, the cervical seal member **210** may be adjustable in one or more directions, orientation, shapes, and/or sizes so as to adjustably contact with the cervix of the patient (and/or surrounding areas around the cervix). A dimension and/or size of the cervical seal member **210** may be greater than an overall size of the cervix so as to seal a uterine cavity of the patient. For example, the cervical seal member **210** may have a diameter of between 50 to 120 mm, or more or less. In some embodiments, the cervical seal member **210** may have a diameter of between 60 to 80 mm. In some embodiments, the cervical seal member **210** may have an outside diameter of about 70 mm.

(89) In example embodiments, the cervical seal member **210** may be formed partially or completely of medical grade rubber, starch, silicone, polymer or copolymer including polyvinyl chloride (PVC), polypropylene (PP), polyethylene (PE), polystyrene (PS), polymethacrylate (PMMA), polyacrylonitrile, or combination thereof. The cervical seal member **210** may be formed with sufficient rigidity so as to maintain its original shape, while also having sufficient flexibility so as to enable the cervical seal member **210** to provide a seal and/or hermetic seal around the cervix of the patient when in operation. The cervical seal member **210** may include elastic and flexible properties so that, when introduced trans-vaginally to contact with a cervix of a patient, the cervical seal member **210** may conform and/or adjusted to fit the patient's body without causing injuries.

(90) The First Anchoring Member (e.g., First Anchoring Member **220***a*).
(91) As illustrated in at least FIGS. **1**A-H, **2**A-D, **3**A-D, **4**A-E, and **4**G, the main assembly **200** includes a first anchoring member (e.g., first anchoring member **220***a*). The first anchoring member **220***a* may be formed on (and/or attached to, formed with, etc.) the elongated body **201** of the main assembly **200**. The first anchoring member **220***a* may be configurable or configured to expand or protrude outwardly away from the elongated body **201** of the main assembly **200**. For example, the first anchoring member **220***a* may be an expandable and/or inflatable member configurable or configured to expand when a positive pressure, gas, liquid, and/or solid is introduced in the first anchoring member **220***a* (e.g., from an external source (not shown)). In an example embodiment, the first anchoring member **220***a* is configurable or configured to provide a vaginal tamponade in at least a portion of the vaginal cavity (or vaginal wall forming the vaginal cavity). The first anchoring member **220***a* may be formed adjacent to the cervical seal member **210**. The first anchoring member **220***a* may be configurable or configured to transition between an anchoring state and a non-anchoring state. In an example embodiment, a default state (or initial state or normal state before anchoring) of the first anchoring member **220***a* is the non-anchoring state.
(92) When transitioning the first anchoring member **220***a* from a non-anchoring state to the anchoring state, the first anchoring member **220***a* may receive a positive pressure, gas, liquid, and/or solid so as to expand and/or protrude to a first overall volume and/or diameter (e.g., expanded outwardly away from the elongated body **201** of the main assembly **200** towards the wall of the vaginal cavity).
(93) When transitioning the first anchoring member **220***a* from the anchoring state to the non-anchoring state, the first anchoring member **220***a* may receive a negative pressure (or not receive a positive pressure, gas, liquid, and/or solid; and/or remove positive pressure, gas, liquid, and/or solid) so as to contract and/or reduce to a second overall volume (e.g., not expanded as outwardly away from the elongated body **201** of the main assembly **200** as compared to the anchoring state). In this regard, the second overall volume is less than the first overall volume.
(94) In another embodiment, a volume, shape, and/or size of the first anchoring member **220***a* can be selectively adjustable in one or more directions. The first anchoring member **220***a* may be in the form of an inflatable or expandable member that can be expanded or inflated with and/or receive and retain a gas, gaseous mixture, pressure, air, water, oil, liquid, semi-solid, etc. In some example embodiments, the first anchoring member **220***a* may be in the form of an inflatable balloon member, expandable member, or the like. The first anchoring member **220***a* may be made of surgical grade materials (e.g., medical grade polyurethane, silicone, polycaprolactone, or the like).
(95) The Second Anchoring Member (e.g., Second Anchoring Member **220***b*).
(96) As illustrated in at least FIGS. **1**A-H, **2**A-D, **3**A-D, **4**A-E, and **4**G, the main assembly **200** includes a second anchoring member (e.g., second anchoring member **220***b*). The second anchoring member **220***b* may be formed on (and/or attached to, formed with, etc.) the elongated body **201** of the main assembly **200**. The second anchoring member **220***b* may be configurable or configured to expand or protrude outwardly away from the elongated body **201** of the main assembly **200**. For example, the second anchoring member **220***b* may be an expandable and/or inflatable member configurable or configured to expand when a positive pressure, gas, liquid, and/or solid is introduced in the first anchoring member **220***a* (e.g., from an external source (not shown)). In an example embodiment, the second anchoring member **220***b* is configurable or configured to provide a vaginal tamponade in at least a portion of the vaginal cavity (or vaginal wall forming the vaginal cavity). The second anchoring member **220***b* may be formed adjacent to the third anchoring member **230** in such a way that the third anchoring member **230** is provided between the first anchoring member **220***a* and the second anchoring member **220***b*. The second anchoring member **220***b* may be configurable or configured to transition between an anchoring state and a non-anchoring state. In an example embodiment, a default state (or initial state or normal state before anchoring) of the second anchoring member **220***b* is the non-anchoring state.

(97) When transitioning the second anchoring member **220***b* from a non-anchoring state to the anchoring state, the second anchoring member **220***b* may receive a positive pressure, gas, liquid, and/or solid so as to and/or protrude to a third overall volume and/or diameter (e.g., expanded outwardly away from the elongated body **201** of the main assembly **200** towards the wall of the vaginal cavity).
(98) When transitioning the second anchoring member **220***b* from the anchoring state to the non-anchoring state, the second anchoring member **220***b* may receive a negative pressure (or not receive a positive pressure, gas, liquid, and/or solid; and/or remove positive pressure, gas, liquid, and/or solid) so as to contract and/or reduce to a fourth overall volume (e.g., not expanded as outwardly away from the elongated body **201** of the main assembly **200** as compared to the anchoring state). In this regard, the fourth overall volume is less than the third overall volume. In example embodiments, the third overall volume may be the same as or similar to the first overall volume. Alternatively or in addition, the fourth overall volume may be the same as or similar to the second overall volume in example embodiments.
(99) In another embodiment, a volume, shape, and/or size of the second anchoring member **220***b* can be selectively adjustable in one or more directions. The second anchoring member **220***b* may be in the form of an inflatable or expandable member that can be inflated with and/or receive and retain a gas, gaseous mixture, pressure, air, water, oil, liquid, semi-solid, etc. In some example embodiments, the second anchoring member **220***b* may be in the form of an inflatable balloon member, expandable member, or the like. The second anchoring member **220***b* may be made of surgical grade materials (e.g., surgical grade polyurethane, silicone, polycaprolactone, or the like).
(100) The Third Anchoring Member (e.g., Third Anchoring Member **230**).
(101) As illustrated in at least FIGS. **1**A-H, **2**A-D, **3**A-D, **4**A-E, and **4**G, the main assembly **200** includes one or more third anchoring members (e.g., third anchoring member **230**). The third anchoring member **230** may be formed or provided on the elongated body **201** of the main assembly **200** between the first and second anchoring members **220***a*, **220***b*. The third anchoring member **230** may be configurable or configured to provide a negative pressure. As described in the present disclosure, when the main assembly **200** is housed in a vaginal cavity of a patient, the third anchoring member **230** is configurable or configured to provide a negative pressure towards at least a portion of the vaginal cavity (or vaginal wall forming the vaginal cavity) in such a way as to collapse, contract, shrink, narrow, and/or condense at least a portion of the vaginal cavity (and/or vaginal wall forming the vaginal cavity) towards the elongated body **201** of the main assembly **200**. In some example embodiments, the third anchoring member **230** may also be configurable or configured to provide positive pressure so as to bring the vaginal cavity (or vaginal wall forming the vaginal cavity) back to its normal shape when needed (e.g., after completing a surgical action, such as treating PPH).
(102) In an example embodiment, the third anchoring member **230** may be configurable or configured to transition between an anchoring state and a non-anchoring state. The anchoring state of the third anchoring member **230** is a state in which the third anchoring member **230** applies at least a first negative pressure. The first negative pressure is an amount of negative pressure required to collapse at least a portion of the vaginal cavity (or vaginal wall forming the vaginal cavity) towards the elongated body **201** of the main assembly **200**. The non-anchoring state of the third anchoring member **230** is a state in which the third anchoring member **230** does not apply at least the first negative pressure (e.g., the third anchoring member **230** does not apply any negative pressure).
(103) In an example embodiment, the elongated body **201** of the main assembly **200** is formed having sufficient rigidity so as to not bend, collapse, or deform when the third anchoring member **230** applies negative pressure (or applies suction). Alternatively or in addition, when the third anchoring member **230** applies negative pressure (e.g., the first negative pressure) and the first and second anchoring members **220***a*, **220***b* are both in the anchoring state, the first and second

anchoring members **220**_a_, **220**_b_ are configured in such a way as to possess sufficient rigidity so as to not collapse or deform towards the third anchoring member **230**.

(104) The Locking Member (e.g., Locking Member **240**).

(105) As illustrated in at least FIG. **1**D, FIG. **1**H, FIG. **7**A, FIG. **7**B, FIG. **7**C, FIG. **8**A, FIG. **8**B, FIG. **8**C, and FIG. **8**D, an example embodiment of the main assembly **200** may include a locking member or mechanism (e.g., locking member **240** or locking mechanism **240**). The locking member **240** may be configurable or configured to transition between a locking state and an unlocking state. When in the locking state, the locking member **240** is configurable or configured to secure, lock, join, restrict, or the like, movement of an element of the hysteroscopic system **100** (e.g., the uterine assembly **300** or the manipulator assembly **400**) relative to the main assembly **200**.

(106) For example, when the uterine assembly **300** is provided through the main channel **202** of the main assembly **200** (e.g., as illustrated in at least FIGS. **6**A, **6**B, **6**C, **6**D and **7**A) and transitioned to the hemostasis state (e.g., the first negative pressure port **310**, uterine expandable member **320**, and second negative pressure port **330** are transitioned from the non-hemostasis state to the hemostasis state when in the uterine cavity of the patient; as illustrated in at least FIGS. **6**F and/or **6**G), the locking member **240** may be transitioned from the unlocking state (e.g., as illustrated in at least FIGS. **7**A, **7**B, **8**B, and **8**D) to the locking state (e.g., as illustrated in at least FIGS. **7**C, **8**A, and **8**C) so as to enable the uterine assembly **300** to be held in the homeostasis state (e.g., as illustrated in at least FIGS. **6**F and/or **6**G) for a prolonged period of time to manage (e.g., stop) the PPH for the patient.

(107) As another example, when the manipulator assembly **400** is provided through the main channel **202** of the main assembly **200** (e.g., as illustrated in at least FIG. **6**I), the locking member **240** may be transitioned from the unlocking state (e.g., as illustrated in at least FIGS. **7**A, **7**B, **8**B, and **8**D) to the locking state (e.g., as illustrated in at least FIGS. **7**C, **8**A, and **8**C) so as to enable the manipulator assembly **400** to be held in position to perform a surgical action in the uterine cavity of the patient.

(108) As illustrated in at least FIGS. **7**A, **7**B, **7**C, **8**A, **8**B, **8**C, and **8**D, an example embodiment of the locking member **240** may include a main locking body **242**. As illustrated in at least FIGS. **7**B and **7**C, the main locking body **242** may be rotatable in one or both of the directions C so as to transition between the unlocking state (e.g., as illustrated in FIG. **7**B) and the locking state (e.g., as illustrated in FIG. **7**C). When the main locking body **242** is rotated to the locking state, one or more lock elements **244** (e.g., as illustrated in FIG. **7**C) are actuated to protrude inwardly so as to contact with and secure, lock, join, restrict, or the like, an element of the hysteroscopic system **100** (e.g., the uterine assembly **300** or the manipulator assembly **400**) relative to the main assembly **200**.

(109) Alternatively or in addition, the locking member **240** may include a locking actuator **243** (e.g., as illustrated in at least FIGS. **8**A, **8**B, **8**C, and **8**D). When the locking actuator **243** is transitioned to the locking state (e.g., as illustrated in at least FIGS. **8**B and **8**D), one or more lock elements **244** (e.g., as illustrated in FIG. **8**C) are actuated to contact with and secure, lock, join, restrict, or the like, an element of the hysteroscopic system **100** (e.g., the uterine assembly **300** or the manipulator assembly **400**) relative to the main assembly **200**.

(110) It is to be understood in the present disclosure that the locking member **240** may be formed in any one or more other shapes, forms, configurations, mechanisms, assemblies, or the like, so long as it secures, locks, joins, restricts, or the like, an element of the hysteroscopic system **100** (e.g., the uterine assembly **300** or the manipulator assembly **400**) relative to the main assembly **200**.

(111) The Vaginal Seal Member (e.g., Vaginal Seal Member **250**).

(112) As illustrated in at least FIG. **2**C and FIG. **2**D, the main assembly **200** includes one or more vaginal seal members (e.g., vaginal seal member **250**). The vaginal seal member **250** may be configurable or configured to properly position the main assembly **200** in a vaginal cavity of the patient. Alternatively or in addition, the vaginal seal member **250** may be configurable or configured to separate, seal, isolate, and/or hermetically seal the vaginal cavity of the patient.

(113) The vaginal seal member **250** may be formed at the in vitro end **200***b* of the main assembly **200**. The vaginal seal member **250** may include a central axis that is coaxial to a central axis of the main assembly **200** (e.g., a central axis of the main channel **202**). The vaginal seal member **250** may be formed along a plane that is orthogonal or substantially orthogonal to the main channel **202** (and/or a central axis formed through the main channel **202**). The vaginal seal member **250** may include an outermost edge portion (or perimeter portion or rim portion) formed between the contact wall **251** and non-contact wall **252** of the vaginal seal member **250**. Such outermost edge portion of the vaginal seal member **250** may be formed along a plane that is orthogonal or substantially orthogonal to the main channel **202** (and/or a central axis formed through the main channel **202**).
(114) To perform the actions, functions, processes, and/or methods described above and in the present disclosure, example embodiments of the vaginal seal member **250** include one or more elements and/or characteristics. For example, the vaginal seal member **250** includes one or more contact walls **251**. In operation, when the main assembly **200** is inserted into the vaginal cavity of the patient, the contact wall **251** is configurable or configured to contact with at least a portion of a vulva (and/or surrounding areas around the vulva) of the patient. In example embodiments, the vaginal seal member **250** may also include one or more non-contact walls **252** (e.g., opposite to the contact wall **251**). In some example embodiments, the vaginal seal member **250** may include one or more sealable members, or the like (not shown), which may be elements that are similar to or the same as the one or more seal members **216** of the cervical seal member **210** (as described in the present disclosure). In some example embodiments, the vaginal seal member **250** may also include one or more suction modules, or the like (not shown), which may be elements that are similar to or the same as the suction modules **218** of the cervical seal member **210** (as described in the present disclosure).
(115) The Contact Wall of the Vaginal Seal Member (e.g., Contact Wall **251**).
(116) In an example embodiment, the vaginal seal member **250** includes one or more contact walls **251**. The contact wall **251** of the vaginal seal member **250** may be configurable or configured to contact with at least a portion of the vulva (and/or surrounding areas around the vulva) of the patient. When contacting with at least a portion of the vulva (and/or surrounding areas around the vulva) of the patient, the contact wall **251** may separate, isolate, seal, and/or hermetically seal the vaginal cavity of the patient (e.g., in cooperation with the sealable member (not shown) of the vaginal seal member **250** and/or one or more other elements of the hysteroscopic system **100**, such as the uterine assembly **300**). The contact wall **251** of the vaginal seal member **250** may include a circular contact portion (and/or any other geometrical shape). The circular contact portion of the contact wall **251** may be formed along a plane that is substantially orthogonal to the first central axis formed through the main channel **202**. The contact wall **251** of the vaginal seal member **250** may include at least a portion that is formed in a concave shape. Alternatively or in addition, the contact wall **251** of the vaginal seal member **250** may include at least a portion that is formed in a convex shape, curve shape, rounded shape, etc. The circular contact portion of the contact wall **251** of the vaginal seal member **250** may be formed along a plane that is orthogonal or substantially orthogonal to the main channel **202** (and/or first central axis formed through the main channel **202**). The contact wall **251** and/or non-contact wall **252** of the vaginal seal member **250** may include one or more flexible portions (e.g., the outermost edge portion formed between the contact wall **251** and non-contact wall **252**). The at least one flexible portion of the contact wall **251** and/or non-contact wall **252** of the vaginal seal member **250** is configured to dynamically adapt to a surface topology of the vulva (and/or surround portions) of the patient (and/or provide a seal and/or hermetic seal).
(117) The Sealable Member of the Vaginal Seal Member (not Shown).
(118) In an example embodiment, the vaginal seal member **250** may also include one or more sealable members (not shown), which may be one or more elements that are similar to, the same as, and/or function similar to and/or the same as the one or more seal members **216** of the cervical seal

member **210** (as described in the present disclosure). The sealable member of the vaginal seal member **250** may be configurable or configured to dynamically adjust or transition between an opening state (e.g., a state in which one or more elements of the hysteroscopic system **100** are provided through the sealable member of the vaginal seal member **250**) and a closed state (e.g., a state in which no elements of the hysteroscopic system **100** are provided through the sealable member of the vaginal seal member **250**).

(119) For example, when a uterine assembly **300** is provided through the sealable member of the vaginal seal member **250** (opening state), the seal member of the vaginal seal member **250** is configurable or configured to provide a hermetic seal around the uterine assembly **300** (or the portion of the uterine assembly **300** provided through the sealable member of the vaginal seal member **250**). The sealable member of the vaginal seal member **250** may be configurable or configured to allow the uterine assembly **300** to slide or pass through in both directions (opening state). As another example, when a manipulator assembly **400** is provided through the sealable member of the vaginal seal member **250** (opening state), the sealable member of the vaginal seal member **250** is configurable or configured to provide a hermetic seal around the manipulator assembly **400** (or the portion of the manipulator assembly **400** provided through the sealable member of the vaginal seal member **250**). The sealable member of the vaginal seal member **250** may be configurable or configured to allow the manipulator assembly **400** to slide or pass through in both directions. The sealable member of the vaginal seal member **250** may be configurable or configured to hermetically seal itself when the manipulator assembly **400** is not provided through the sealable member of the vaginal seal member **250** (closed state). The sealable member of the vaginal seal member **250** may be formed coaxial to the vaginal seal member **250** (and/or a central axis formed through the vaginal seal member **250**). Alternatively or in addition, the sealable member of the vaginal seal member **250** may be formed coaxial to the main channel **202** (and/or a central axis formed through the main channel **202**).

(120) The Suction Modules of the Vaginal Seal Member (not Shown).

(121) In an example embodiment, the vaginal seal member **250** may also include one or more suction modules (not shown). The one or more suction modules of the vaginal seal member **250** may be formed on the contact wall **251** of the vaginal seal member **250**. The one or more suction modules of the vaginal seal member **250** may be configurable or configured to provide a negative pressure. When in operation, such negative pressure may enable or facilitate a seal and/or hermetic seal between the vaginal seal member **250** and a vulva of the patient (and/or surrounding area of the vulva of the patient).

(122) In another example embodiment, the vaginal seal member **250** may be adjustable in one or more directions and/or orientation so as to contact with the vulva of the patient (and/or surrounding areas of the vulva of the patient). A dimension and/or size of the vaginal seal member **250** may be greater than an overall size of the vulva of the patient so as to seal the vulva (and/or surrounding areas) of the patient. For example, the vaginal seal member **250** may have a diameter of between 50 to 120 mm, or more or less. In some embodiments, the vaginal seal member **250** may have a diameter of between 60 to 80 mm. In some embodiments, the vaginal seal member **250** may have an outside diameter of about 70 mm.

(123) In example embodiments, the vaginal seal member **250** may be formed partially or completely of medical grade rubber, starch, silicone, polymer or copolymer including polyvinyl chloride (PVC), polypropylene (PP), polyethylene (PE), polystyrene (PS), polymethacrylate (PMMA), polyacrylonitrile, or combination thereof. The vaginal seal member **250** may be formed with sufficient rigidity so as to maintain its original shape, while also having sufficient flexibility so as to enable the vaginal seal member **250** to provide a seal and/or hermetic seal around the vulva of the patient when in operation. The vaginal seal member **250** may include elastic and flexible properties so that, when in use, the vaginal seal member **250** may conform and/or adjusted to fit an exterior of the patient's body without causing injuries.

(124) The First Extendible Member (e.g., First Extendible Member **260**).
(125) As illustrated in at least FIG. **3**B, the main assembly **200** may include one or more first extendible members (e.g., first extendible member **260**). The first extendible member **260** may be configurable or configured to selectively adjust or change a length of the main assembly **200** (and/or a distance between the in vivo end **200***a* and the in vitro end **200***b* of the main assembly **200**).
(126) For example, the first extendible member **260** may be configurable or configured to extend or contract/shorten in Direction A (e.g., as illustrated in at least FIG. **3**B) to adjust the overall length of the main assembly **200**. The first extendible member **260** may be formed on, in, and/or as a part of the main assembly **200** (e.g. adjacent to the second anchoring member **220***b* and/or at any other location along the main assembly **200**). When the main assembly **200** is introduced trans-vaginally, the first extendible member **260** may be adjusted such that each element of the main assembly **200** (e.g. the cervical seal member **210**, the first anchoring member **220***a*, the second anchoring member **220***b*, the third anchoring member **230**, and vaginal seal member **250**, etc.) is in a patient-specific position. Alternatively or in addition, the first extendible member **260** may be adjusted to suit specific anatomy (e.g., vaginal cavity dimensions) of each patient.
(127) In example embodiments, the first extendible member **260** may be formed in any shape, form, size, and/or dimension (e.g., tubular shaped member, cylindrical shaped member, hollow shaped member, square tube shaped member, etc.).
(128) The Uterine Assembly (e.g., Uterine Assembly **300**).
(129) As illustrated in at least in FIGS. **1**A-**1**H and **2**A-**2**D, the hysteroscopic system **100** includes a uterine assembly (e.g., uterine assembly **300**). The uterine assembly **300** may be configurable or configured to be inserted through (and/or slide through) the main channel **202** of the main assembly **200** from the in vitro end **200***b* of the main assembly **200** until the in vivo end **300***a* of the uterine assembly **300** passes through the main channel **202** of the main assembly **200** and into the uterine cavity of the patient. In doing so, the uterine assembly **300** may be inserted through the sealable member **216** of the cervical seal member **210**. In example embodiments in which the vaginal seal member **250** includes a sealable member (not shown; as described in the present disclosure), the uterine assembly **300** may also be inserted through the sealable member of the vaginal seal member **250**.
(130) To perform the actions, functions, processes, and/or methods described above and in the present disclosure, the uterine assembly **300** may include one or more elements.
(131) For example, the uterine assembly **300** includes and/or is formed as an elongated body **301** having an in vivo end **300***a* and an in vitro end **300***b* opposite to the in vivo end **300***a*. The elongated body **301** of the uterine assembly **300** may be formed in one or more of a variety of shapes, dimensions, configurations, etc. For example, the elongated body **301** of the uterine assembly **300** may be formed in the shape of a cylindrical tube, or the like, and/or with a circular, elliptical, oval, or other cross sectional shape. The elongated body **301** of the uterine assembly **300** may include and/or be formed having a central axis.
(132) The uterine assembly **300** also includes one or more first negative pressure ports **310**. The uterine assembly **300** also includes one or more uterine expandable members **320**. The uterine assembly **300** also includes one or more second negative pressure ports **330**. The uterine assembly **300** may also include one or more locking members or mechanisms (not shown), which may be used (either alone or in cooperation with locking member **240** of the main assembly **200**; such locking member of the uterine assembly **300** being similar to that of locking member **240**) to secure, lock, join, restrict, or the like, movement of the uterine assembly **300** relative to the main assembly **200**. The uterine assembly **300** may also include one or more bendable members **370** (e.g., as illustrated in at least FIG. **5**C). The uterine assembly **300** may also include one or more second extendible members **340** (as illustrated in at least FIG. **5**D).
(133) These and other elements of the uterine assembly **300** will now be further described with

reference to the accompanying figures.
(134) The Elongated Body of the Uterine Assembly (e.g., Elongated Body **301**).
(135) As illustrated in at least FIGS. **1**A-**1**H, **2**A-**2**D, and **5**A-D, the uterine assembly **300** includes an elongated body of uterine assembly (e.g., elongated body **301**). The elongated body **301** of the uterine assembly **300** may function as a core structure to provide structural rigidity for the uterine assembly **300**, and to secure, lock, join, restrict, or the like, a position of the uterine assembly **300** relative to the main assembly **200***a*. The elongated body **301** of the uterine assembly **300** may be configurable or configured to support each element of the uterine assembly **300** (e.g., the first negative pressure port **310**, the uterine expandable member **320**, the second negative pressure port **330**, the bendable member **370**, the extendible member **340**, etc.). The elongated body **301** of the uterine assembly **300** is formed having sufficient rigidity so as to not bend, collapse, or deform when in operation, including when the first negative pressure port **310** is in the hemostasis state; when the uterine expandable member **320** is in the hemostasis state; when the second negative pressure port **330** is in the hemostasis state; when the bendable member **370** is selectively configured to bend in one or more locations along the bendable member **370** and/or bend in one or more directions and/or angles; when the extendible member **340** is selectively configured to extend or contract in length; and/or when the uterine assembly **300** is secured, locked, joined, restricted, or the like, relative to the main assembly **200**.
(136) The elongated body **301** of the uterine assembly **220** may be formed as a tubular shaped member, cylindrical shaped member, hollow member, square tubular shaped member, etc.
(137) The First Negative Pressure Port (e.g., First Negative Pressure Port **310**).
(138) As illustrated in at least FIGS. **1**A-**1**H, **2**A-**2**D, and **5**A-D, the uterine assembly **300** includes one or more first negative pressure ports (e.g., first negative pressure port **310**). The first negative pressure port **310** may be formed or provided on a distal-most end (or tip) of the in vivo end **300***a* of the elongated body **301** of the uterine assembly **300** (e.g., see FIGS. **1**A-H and **2**A-D). Alternatively or in addition, the first negative pressure port **310** may be formed or provided on a side of the in vivo end **300***a* of the elongated body **301** of the uterine assembly **300** (e.g., see FIGS. **1**B and **1**F). The first negative pressure port **310** may be configurable or configured to provide a negative pressure. As described in the present disclosure, when the uterine assembly **300** is housed in a uterine cavity of a patient, the first negative pressure port **310** is configurable or configured to provide a negative pressure towards at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) in such a way as to collapse, contract, shrink, narrow, and/or condense at least a portion of the uterine cavity (and/or uterine wall forming the uterine cavity) towards the elongated body **301** of the uterine assembly **300**. In some example embodiments, the first negative pressure port **310** may also be configurable or configured to provide positive pressure so as to bring the uterine cavity (or uterine wall forming the uterine cavity) back to its normal (or near-normal) shape when needed (e.g., after completing a surgical action, such as treating PPH).
(139) In an example embodiment, the first negative pressure port **310** may be configurable or configured to transition between a hemostasis state (or anchoring state) and a non-hemostasis state (or non-anchoring state). The hemostasis state of the first negative pressure port **310** is a state in which the first negative pressure port **310** applies at least a second negative pressure. The second negative pressure is an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body **301** of the uterine assembly **300**. The non-hemostasis state of the first negative pressure port **310** is a state in which the first negative pressure port **310** does not apply at least the second negative pressure (e.g., the first negative pressure port **310** does not apply any negative pressure).
(140) In an example embodiment, the elongated body **301** of the uterine assembly **300** is formed having sufficient rigidity so as to not bend, collapse, or deform when the first negative pressure port **310** applies negative pressure (or applies suction). Alternatively or in addition, when the first negative pressure port **310** applies negative pressure (e.g., the second negative pressure) and the

uterine expandable member **320** and second negative pressure port **330** are both in the hemostasis state, the uterine expandable member **320** is configured in such a way as to possess sufficient rigidity so as to not collapse or deform.

(141) The first negative pressure port **310** may be configurable or configured to provide negative pressure via a negative pressure source (not shown). The first negative pressure port **310** may also be configurable or configured to decrease a volume of the uterine cavity or cervix of the patient by providing a vacuum aspiration by any mechanically suitable methods. The first negative pressure port **310** may be in a form of a port, opening, groove, hole, aperture, orifice, slot, vent, or the like so as to allow the negative pressure to be introduced to the uterine cavity or cervix of the patient.

(142) The Uterine Expandable Member (e.g., Uterine Expandable Member **320**).

(143) As illustrated in at least FIGS. **1**A-H, **2**A-D, and **5**A-D, the uterine assembly **300** includes one or more uterine expandable members (e.g., uterine expandable member **320**). The uterine expandable member **320** may be formed on (and/or attached to, formed with, etc.) the elongated body **301** of the uterine assembly **300**. The uterine expandable member **320** may be configurable or configured to expand or protrude outwardly away from the elongated body **301** of the uterine assembly **300**. For example, the uterine expandable member **320** may be an expandable and/or inflatable member configurable or configured to expand when a positive pressure, gas, liquid, and/or solid is introduced in the uterine expandable member **320** (e.g., from an external source (not shown)). The uterine expandable member **320** may be formed adjacent to the first negative pressure port **310**. In example embodiments, the uterine expandable member **320** may be formed between the first negative pressure port **310** and the second negative pressure port **330**. The uterine expandable member **320** may be configurable or configured to transition between a hemostasis state (or anchoring state) and a non-hemostasis state (or non-anchoring state). In an example embodiment, a default state (or initial state or normal state before performing hemostasis in the uterine cavity) of the uterine expandable member **320** is the non-hemostasis state.

(144) When transitioning the uterine expandable member **320** from a non-hemostasis state to the hemostasis state, the uterine expandable member **320** may receive a positive pressure (or gas, liquid, and/or solid) so as to expand to a third overall volume (e.g., expanded outwardly away from the elongated body **301** of the uterine assembly **300** towards the wall of the uterine cavity).

(145) When transitioning the uterine expandable member **320** from the hemostasis state to the non-hemostasis state, the uterine expandable member **320** may receive a negative pressure (or not receive a positive pressure, gas, liquid, and/or solid; and/or remove positive pressure, gas, liquid, and/or solid) so as to contract and/or reduce to a fourth overall volume (e.g., not expanded as outwardly away from the elongated body **301** of the uterine assembly **300** as compared to the hemostasis state). In this regard, the fourth overall volume is less than the third overall volume.

(146) In another embodiment, a volume, shape, and/or size of the uterine expandable member **320** can be selectively adjustable in one or more directions. The uterine expandable member **320** may be in the form of an inflatable or expandable object that can be expanded or inflated with and/or receive and retain a gas, gaseous mixture, pressure, air, water, oil, liquid, semi-solid, etc. In some example embodiments, the uterine expandable member **320** may be in the form of an inflatable balloon member, expandable member, or the like. The uterine expandable member **320** may be made of surgical grade materials (e.g., surgical grade polyurethane, silicone, polycaprolactone, or the like).

(147) The Second Negative Pressure Port (e.g., Second Negative Pressure Port **330**).

(148) As illustrated in at least FIGS. **1**B, IF, **2**A-D, and **5**A-D, the uterine assembly **300** includes one or more second negative pressure ports (e.g., second negative pressure port **330**). The second negative pressure port **330** may be formed or provided on the elongated body **301** of the uterine assembly **300**. For example, the second negative pressure port **330** may be formed or provided on the elongated body **301** of the uterine assembly **300** in such a way that the uterine expandable member **320** is provided between the first negative pressure port **310** and the second negative

pressure port **330**. The second negative pressure port **330** may be configurable or configured to provide a negative pressure. As described in the present disclosure, when the uterine assembly **300** is housed in a uterine cavity of a patient, the second negative pressure port **330** is configurable or configured to provide a negative pressure towards at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) in such a way as to collapse, contract, shrink, narrow, and/or condense at least a portion of the uterine cavity (and/or uterine wall forming the uterine cavity) towards the elongated body **301** of the uterine assembly **300**. In some example embodiments, the second negative pressure port **330** may also be configurable or configured to provide positive pressure so as to bring the uterine cavity (or uterine wall forming the uterine cavity) back to its normal (or near-normal) shape when needed (e.g., after completing a surgical action, such as treating PPH).

(149) In an example embodiment, the second negative pressure port **330** may be configurable or configured to transition between a hemostasis state (or anchoring state) and a non-hemostasis state (or non-anchoring state). The hemostasis state of the second negative pressure port **330** is a state in which the second negative pressure port **330** applies at least a third negative pressure. The third negative pressure is an amount of negative pressure required to collapse at least a portion of the uterine cavity (or uterine wall forming the uterine cavity) towards the elongated body **301** of the uterine assembly **300**. The non-hemostasis state of the second negative pressure port **330** is a state in which the second negative pressure port **330** does not apply at least the third negative pressure (e.g., the second negative pressure port **330** does not apply any negative pressure).

(150) In example embodiments, the first negative pressure port **310** and second negative pressure port **330** are configurable or configured to cooperate to provide a sufficient collective negative pressure (e.g., the second negative pressure plus the third negative pressure) to collapse, contract, shrink, narrow, and/or condense most or all of the uterine cavity (and/or uterine wall forming the uterine cavity) towards the uterine expandable member **320** (when in the hemostasis state).

(151) In an example embodiment, the elongated body **301** of the uterine assembly **300** is formed having sufficient rigidity so as to not bend, collapse, or deform when the second negative pressure port **330** applies negative pressure (or applies suction). Alternatively or in addition, when the second negative pressure port **330** applies negative pressure (e.g., the third negative pressure) and the uterine expandable member **320** and first negative pressure port **310** are both in the hemostasis state, the uterine expandable member **320** is configured in such a way as to possess sufficient rigidity so as to not collapse or deform.

(152) The second negative pressure port **330** may be configurable or configured to provide negative pressure via a negative pressure source (not shown). The second negative pressure port **330** may also be configurable or configured to decrease a volume of the uterine cavity or cervix of the patient by providing a vacuum aspiration by any mechanically suitable methods. The second negative pressure port **330** may be in a form of a port, opening, groove, hole, aperture, orifice, slot, vent, or the like so as to allow the negative pressure to be introduced to the uterine cavity or cervix of the patient.

(153) The Bendable Member (e.g., Bendable Member **370**).

(154) As illustrated in at least FIG. **5**C, the uterine assembly **300** includes one or more bendable members (e.g., bendable member **370**). The bendable member **370** may be configurable or configured to facilitate a bending of the in vivo end **300***a* of the uterine assembly **300** at one or more locations and in one or more directions. The bendable member **370** may be selectively adjusted to bend in multiple angles, directions, and locations along the bendable member **370** so as to adjustably direct as needed the vivo end **300***a* of the uterine assembly **300** relative to the patient's uterine cavity.

(155) In example embodiments, the bendable member **370** may be formed adjacent to the second negative pressure port **330** (e.g., between the second negative pressure port **330** and the in vitro end **300***b* of the uterine assembly **300**). Alternatively or in addition, the bendable member **370** may be

formed between the uterine expandable member **320** and the second negative pressure port **330**. Alternatively or in addition, the bendable member **370** may be formed between the first negative pressure port **310** and the uterine expandable member **320**. In example embodiments, the bendable member **370** may be configurable or configured to connect the in vivo end **300***a* of the uterine assembly **300** to the elongated body **301** (and/or in vitro end **300***b*) of uterine assembly **300**.

(156) In example embodiments, the bendable member **370** may be formed using any medical grade flexible, elastic, plastic, rigid, deformable, etc. material so long as the bendable member **370** can be selectively adjusted to bend in one or more bending angles, directions, and/or locations while rigid enough to maintain the vivo end **300***a* of the uterine assembly **300** in a desired direction, position, orientation, and/or location.

(157) The Second Extendible Member (e.g., Second Extendible Member **340**).

(158) As illustrated in at least FIG. **5**D, the uterine assembly **300** may include one or more second extendible members (e.g., second extendible member **340**). The second extendible member **340** may be configurable or configured to selectively adjust or change a length of the uterine assembly **300** (and/or a distance between the in vivo end **300***a* and the in vitro end **300***b* of the uterine assembly **300**).

(159) For example, the second extendible member **340** may be configurable or configured to extend or contract/shorten in Direction B (e.g., as illustrated in at least FIG. **5**D) to adjust the overall length of the uterine assembly **300**. The second extendible member **340** may be formed on, in, and/or as a part of the uterine assembly **300** (e.g. adjacent to the uterine expandable member **320**, between the uterine expandable member **320** and the second negative pressure port **330**, and/or at any other location along the uterine assembly **300**). When the uterine assembly **300** is inserted into the uterine cavity of the patient, the second extendible member **340** may be adjusted such that each element of the uterine assembly **300** (e.g., the first negative pressure port **310**, the uterine expandable member **320**, the second negative pressure port **330**, etc.) is in a patient-specific position. Alternatively or in addition, the second extendible member **340** may be adjusted to suit specific anatomy (e.g., uterine cavity dimensions) of each patient.

(160) In example embodiments, the second extendible member **340** may be formed in any shape, form, size, and/or dimension (e.g., tubular shaped member, cylindrical shaped member, hollow shaped member, square tube shaped member, etc.).

(161) The Manipulator Assembly (e.g., Manipulator Assembly **400**).

(162) As illustrated in at least FIG. **6**I, an example embodiment of the hysteroscopic system **100** may further include a manipulator assembly (e.g., manipulator assembly **400**). The manipulator assembly **400** may be configurable or configured to be inserted through and/or slidably housed in the main channel **202** of the main assembly **200**.

(163) The manipulator assembly **400** may be formed as an elongated body **401** having an in vivo end **400***a* and an in vitro end **400***b*. The manipulator assembly **400** may be inserted to mobilize in different positions of the uterine cavity. The manipulator assembly **400** includes one or more end effectors **410** for use in performing a surgical action (e.g., on one or more parts of the uterine cavity or uterine wall forming the uterine cavity of the patient).

(164) The manipulator assembly **400** may also include one or more bendable sections **420**. The bendable section **420** may be configurable or configured to facilitate a bending of the in vivo end **400***a* of the manipulator assembly **400** at one or more locations and in one or more directions. The bendable section **420** may be selectively adjusted to bend in multiple angles, directions, and locations along the bendable section **420** so as to adjustably direct as needed the end effector **410** relative to the patient's uterine cavity.

(165) In example embodiments, the manipulator assembly **400** may include one or more extendible sections (not shown). The extendible section of the manipulator assembly **400** may be configurable or configured to selectively adjust or change a length of the manipulator assembly **400** (and/or a distance between the in vivo end **400***a* and the in vitro end **400***b* of the manipulator assembly **400**).

(166) The Controller (not Shown).
(167) In an example embodiment, the hysteroscopic system **100** includes one or more controllers (not shown). The controller is configurable or configured to control and/or manage one or more elements of the hysteroscopic system **100** (as described in the present disclosure). The controller may perform such controlling and/or managing of one or more elements of the hysteroscopic system **100** by receiving instructions, commands, and/or actions from onsite and/or remote one or more surgeons or operators. Alternatively or in addition, the controller may perform such controlling and/or managing of one or more elements of the hysteroscopic system **100**, in part or in whole, via artificial intelligence, or the like.
(168) For example, the controller is configurable or configured to manage and/or control one or more elements of the main assembly **200**. More specifically, the controller may be configurable or configured to transition the first anchoring member **220***a* between the anchoring state and the non-anchoring state. The controller may also be configurable or configured to transition the second anchoring member **220***b* between the anchoring state and the non-anchoring state. The controller may also be configurable or configured to transition the third anchoring member **230** between the anchoring state and the non-anchoring state. The controller may also be configurable or configured to transition the main assembly **200** between the anchoring state and the non-anchoring state. The controller may also be configurable or configured to actuate the extendible section **260** to extend and contract so as to change a length of the main assembly **200**. The controller may also be configurable or configured to transition the locking member **240** between the locking state and the unlocking state. The controller may also be configurable or configured to cause the cervical seal member **210** to seal or hermetically seal the uterine cavity of the patient. The controller may also be configurable or configured to cause the vaginal seal member **250** to seal or hermetically seal the vaginal cavity of the patient. The controller may also be configurable or configured to position the main assembly **200** into the vaginal cavity of the patient, including removing the main assembly **200** from the vaginal cavity of the patient.
(169) As another example, the controller is configurable or configured to manage and/or control one or more elements of the uterine assembly **300**. More specifically, the controller may be configurable or configured to transition the first negative pressure port **310** between the hemostasis state and the non-hemostasis state. The controller may also be configurable or configured to transition the uterine expandable member **320** between the hemostasis state and the non-hemostasis state. The controller may also be configurable or configured to transition the second negative pressure port **330** between the hemostasis state and the non-hemostasis state. The controller may also be configurable or configured to transition the uterine assembly **300** between the hemostasis state and the non-hemostasis state. The controller may also be configurable or configured to actuate the bendable section **370** to bend in a plurality of directions, curvatures, and/or locations. The controller may also be configurable or configured to actuate the extendible section **340** to extend and contract so as to change a length of the uterine assembly **300**. The controller may also be configurable or configured to transition the locking member of the uterine assembly between the locking state and the unlocking state. The controller may also be configurable or configured to position the in vivo end **300***a* of the uterine assembly **300** into the uterine cavity of the patient, including removing the uterine assembly **300** from the uterine cavity of the patient.
(170) In yet another example, the controller is configurable or configured to manage and/or control one or more elements of the manipulator assembly **400**. More specifically, the controller may be configurable or configured to actuate the end effector **410** to perform a surgical action. The controller may also be configurable or configured to actuate the bendable section **420** to bend in a plurality of directions, curvatures, and/or locations. The controller may also be configurable or configured to actuate an extendible section of the manipulator assembly **400** to extend and contract so as to change a length of the manipulator assembly **400**. The controller may also be configurable or configured to position the in vivo end **400***a* of the manipulator assembly **400** into the uterine

cavity of the patient, including removing the manipulator assembly **400** from the uterine cavity of the patient.
(171) As used in the present disclosure, when applicable, the controller (and/or one or more of its elements) may include one or more computing devices, processors, servers, systems, cloud-based computing, or the like, and/or functionality of one or more processors, computing devices, servers, systems, cloud-based computing, or the like. The controller (and/or one or more of its elements) may be or have any processor, server, system, device, computing device, controller, microprocessor, microcontroller, microchip, semiconductor device, or the like, configurable or configured to perform one or more of the actions described above and in the present disclosure. Alternatively or in addition, the controller (and/or one or more of its elements) may include and/or be a part of a virtual machine, processor, computer, node, instance, host, or machine, including those in a networked computing environment.
(172) Example Embodiments of a Method of Managing Post-Partum Hemorrhaging and/or a Method of Configuring a Hysteroscopic System to Manage Post-Partum Hemorrhaging.
(173) As illustrated in at least FIGS. **6**A-I, an example embodiment of a method of managing post-partum hemorrhaging (PPH) and/or a method of configuring a hysteroscopic system (e.g., hysteroscopic system **100**) to manage PPH includes one or more actions performed using one or more elements of the hysteroscopic system **100**.
(174) For example, the method may include configuring a main assembly **200** to be in a non-anchoring state (as described in the present disclosure), if the main assembly **200** is not already in the non-anchoring state. The method may also include configuring a uterine assembly **300** to be in a non-hemostasis state (as described in the present disclosure), if the uterine assembly **300** is not already in the non-hemostasis state.
(175) The method further includes configuring the uterine assembly **300** and main assembly **200** to be in a first insertion configuration by inserting the uterine assembly **300** into a main channel **202** of the main assembly **200** in such a way that at least a portion of the uterine assembly **300** is housed in the main channel **202** of the main assembly **200** (e.g., as illustrated in at least FIG. **6**A). While in the first insertion configuration, at least a portion of an in vivo end **300***a* of the uterine assembly **300** may not be housed in the main channel **202**. In example embodiments, the first insertion configuration may include transitioning a locking member **240** (and/or locking member of the uterine assembly **300**, not shown) to a locking state so as to secure or lock the uterine assembly **300** relative to the main assembly **200** (as described in the present disclosure). It is to be understood that the locking member **240** (and/or locking member of the uterine assembly **300**, not shown) may be transitioned to the locking state before and/or after both main assembly **200** and uterine assembly **300** are inserted into a vaginal cavity of the patient (and the in vivo end **300***a* of the uterine assembly is inserted into a uterine cavity of the patient). As illustrated in at least FIGS. **6**A and **6**B, the uterine assembly **300** and the main assembly **200** (while in the first insertion configuration) may then be introduced trans-vaginally into the vaginal cavity of the patient (and the in vivo end **300***a* of the uterine assembly introduced into a uterine cavity of the patient) until a cervical seal member **210**, which is formed at the in vivo end **200***a* of the main assembly **200**, is brought into contact with at least a portion of a cervix (and/or areas surrounding the cervix) of the patient (as illustrated in at least FIG. **6**B).
(176) Alternative to the first insertion configuration, the method may include first inserting the main assembly **200** into the vaginal cavity of the patient until the cervical seal member **210** is brought into contact with at least a portion of the cervix (and/or areas surrounding the cervix) of the patient. Thereafter, the method may include inserting the uterine assembly **300** through the main channel **202** of the main assembly **200** until the in vivo end **300***a* of the uterine assembly **300** passes through the main channel **202** (and is inserted into the uterine cavity of the patient, as illustrated in at least FIG. **6**B). In example embodiments having the sealable member **216** sealable member for the vaginal seal member **250**, the inserting of the uterine assembly **300** through the

main channel **202** further includes inserting the in vivo end **300***a* of the uterine assembly **300** through the sealable member **216** and the sealable member of the vaginal seal member **250**.
(177) After the main assembly **200** is housed in the vaginal cavity of the patient and the in vivo end **300***a* of the uterine assembly **300** is housed in the uterine cavity of the patient (as illustrated in at least FIG. **6**B), the method may include sealing or hermetically sealing the uterine cavity from the vaginal cavity of the patient via the cervical seal member **210**. Alternatively or in addition, the cervical seal member **210** is configured to ensure that the main assembly **200** is properly positioned in the vaginal cavity of the patient, and to ensure that the main assembly **200** is not inserted into the uterine cavity of the patient.
(178) The method may include transitioning the main assembly **200** to be in the anchoring state (as described in the present disclosure). As illustrated in at least FIG. **6**C, this may include first transitioning the first anchoring member **220***a* to be in the anchoring state and transitioning the second anchoring member **220***b* to be in the anchoring state. As illustrated in at least FIG. **6**D, the third anchoring member **230** may then be transitioned to be in the anchoring state, which enables or causes at least a portion of the vaginal cavity of the patient (and/or vaginal walls forming the vaginal cavity) to collapse towards the elongated body **201** of the main assembly **200** (i.e., towards the third anchoring member **230**). In example embodiments, the first anchoring member **220***a*, second anchoring member **220***b*, and third anchoring member **230** are all transitioned to the anchoring state (and in the above order, or in any order).
(179) After the main assembly **200** is transitioned to the anchoring state, the method includes transitioning the uterine assembly **300** to be in the hemostasis state (as described in the present disclosure). As illustrated in at least FIG. **6**E, this may include transitioning the uterine expandable member **320** to be in the hemostasis state, transitioning the first negative pressure port **310** to be in the hemostasis state, and/or transitioning the second negative pressure port **330** to be in the hemostasis state. In example embodiments, the first negative pressure port **310**, uterine expandable member **320**, and second negative pressure port **330** are all transitioned to the hemostasis state. As illustrated in at least FIG. **6**F, transitioning the uterine assembly **300** to be in the hemostasis state enables or causes at least a portion of the uterine cavity of the patient (and/or uterine wall forming the uterine cavity) to collapse towards the elongated body **301** of the uterine assembly **300** (i.e., towards the first negative pressure port **310**, uterine expandable member **320**, and second negative pressure port **330**). In example embodiments, most or all of the uterine cavity of the patient (and/or uterine wall forming the uterine cavity) collapses towards and onto (or enveloping) at least the uterine expandable member **320** when the uterine assembly **300** is transitioned to the hemostasis state.
(180) After the uterine assembly **300** is transitioned to the hemostasis state and the main assembly **200** is transitioned to the anchoring state, the method may include pulling or sliding the uterine assembly **300** (which remains in the hemostasis state) relative to the main assembly **200** (which remains in the anchoring state), as illustrated in at least FIG. **6**G, so as to further manage or treat PPH (e.g., performing an accordioning of the uterine cavity of the patient (and/or uterine walls forming the uterine cavity)). In example embodiments in which the locking member **240** (and/or locking member of the uterine assembly **300**) is provided and in the locking state, the method includes first transitioning the locking member **240** to the unlocking state.
(181) The method further includes transitioning the uterine assembly **300** from the hemostasis state to the non-hemostasis state, including transitioning the uterine expandable member **320** to be in the non-hemostasis state, transitioning the first negative pressure port **310** to be in the non-hemostasis state, and/or transitioning the second negative pressure port **330** to be in the non-hemostasis state. As illustrated in at least FIG. **6**H, the uterine assembly **300** is then pulled, removed, or withdrawn from the uterine cavity of the patient and the main channel **202** of the main assembly **200**.
(182) It is recognized in the present disclosure that after the uterine assembly **300** is transitioned to the non-hemostasis state, removed from the uterine cavity of the patient, and removed from the

main channel **202** of the main assembly **200**, the uterine cavity of the patient may remain in the same or similar shape as when the uterine assembly **300** was in the hemostasis state. In this regard, the method may include inserting a manipulator assembly **400** through the main channel **202** of the main assembly **200** in such a way that the end effector **410** of the manipulator assembly **400** can perform a surgical action in the uterine cavity of the patient (which can be properly positioned via bendable section **420**, extendible section (not shown)). Although not shown, the manipulator assembly **400** may also include one or more imaging assemblies for use in enabling a surgeon to view inside the uterine cavity of the patient so as to perform the surgical actions with the end effector **410**.

(183) While various embodiments in accordance with the disclosed principles have been described above, it should be understood that they have been presented by way of example only, and are not limiting. Thus, the breadth and scope of the example embodiments described in the present disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the claims and their equivalents issuing from this disclosure. Furthermore, the above advantages and features are provided in described embodiments, but shall not limit the application of such issued claims to processes and structures accomplishing any or all of the above advantages.

(184) For example, "communication," "communicate," "connection," "connect," or other similar terms should generally be construed broadly to mean a wired, wireless, and/or other form of, as applicable, connection between elements, devices, computing devices, telephones, processors, controllers, servers, networks, telephone networks, the cloud, and/or the like, which enable voice and/or data to be sent, transmitted, broadcasted, received, intercepted, acquired, and/or transferred (each as applicable).

(185) As another example, "seal", "sealing", "sealed", "sealable", or the like as referred herein should generally be construed broadly to mean to any actions including to close, cover, attach, fix, adhere, hinder, block, or obstruct and/or the like, which partly or substantially prevent, limit, interfere the activity or progress of the air or liquid, etc. to pass in or out.

(186) As another example, "in vivo end" as referred herein should generally be construed broadly to mean a part that is inside an organism and more specifically to the part that is in or on the living tissue of a whole, living multicellular organism (animal), such as a mammal.

(187) As another example, "in vitro end" as referred herein should generally be construed broadly to mean a part that is outside an organism and more specifically to the part that is not in or on the living tissue of a whole, living multicellular organism (animal), such as a mammal.

(188) Various terms used herein have special meanings within the present technical field. Whether a particular term should be construed as such a "term of art" depends on the context in which that term is used. Such terms are to be construed in light of the context in which they are used in the present disclosure and as one of ordinary skill in the art would understand those terms in the disclosed context. The above definitions are not exclusive of other meanings that might be imparted to those terms based on the disclosed context.

(189) Words of comparison, measurement, and timing such as "at the time," "equivalent," "during," "complete," and the like should be understood to mean "substantially at the time," "substantially equivalent," "substantially during," "substantially complete," etc., where "substantially" means that such comparisons, measurements, and timings are practicable to accomplish the implicitly or expressly stated desired result.

(190) Additionally, the section headings and topic headings herein are provided for consistency with the suggestions under various patent regulations and practice, or otherwise to provide organizational cues. These headings shall not limit or characterize the embodiments set out in any claims that may issue from this disclosure. Specifically, a description of a technology in the "Background" is not to be construed as an admission that technology is prior art to any embodiments in this disclosure. Furthermore, any reference in this disclosure to "invention" in the

singular should not be used to argue that there is only a single point of novelty in this disclosure. Multiple inventions may be set forth according to the limitations of the claims issuing from this disclosure, and such claims accordingly define the invention(s), and their equivalents, that are protected thereby. In all instances, the scope of such claims shall be considered on their own merits in light of this disclosure, but should not be constrained by the headings herein.

Claims

1. A hysteroscopic system for managing post-partum hemorrhaging, the hysteroscopic system comprising: a main assembly, the main assembly formed as an elongated body having an in vivo end and an in vitro end, wherein, when in operation, the in vivo end of the main assembly is configured to be inserted into and housed in a vaginal cavity of a patient and the in vitro end of the main assembly is configured to remain outside of the vaginal cavity of the patient, the main assembly including: a main channel formed through the elongated body of the main assembly between the in vivo and in vitro ends of the main assembly, the main channel formed along a first central axis; a cervical seal member, the cervical seal member formed at the in vivo end of the main assembly and having a central axis coaxial to the first central axis, the cervical seal member having a contact wall and a non-contact wall opposite to the contact wall of the cervical seal member, the contact wall of the cervical seal member configured to contact with at least a portion of a cervix of the patient, the cervical seal member configured to hermetically isolate a uterine cavity of the patient from the vaginal cavity of the patient when: the contact wall of the cervical seal member is positioned to be in contact with at least a portion of the cervix of the patient; a first anchoring member, the first anchoring member formed on the elongated body of the main assembly and adjacent to the cervical seal member, the first anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state of the first anchoring member being a state in which the first anchoring member has a first overall volume, the non-anchoring state of the first anchoring member being a state in which the first anchoring member has a second overall volume, the first overall volume greater than the second overall volume; a second anchoring member, the second anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state of the second anchoring member being a state in which the second anchoring member has a third overall volume, the non-anchoring state of the second anchoring member being a state in which the second anchoring member has a fourth overall volume, the third overall volume greater than the fourth overall volume; and a third anchoring member, the third anchoring member formed between the first and second anchoring members, the third anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state of the third anchoring member being a state in which the third anchoring member applies at least a first negative pressure, the first negative pressure being an amount of negative pressure required to collapse at least a portion of the vaginal cavity towards the elongated body of the main assembly, the non-anchoring state of the third anchoring member being a state in which the third anchoring member does not apply at least the first negative pressure; and a uterine assembly, the uterine assembly formed as an elongated body having an in vivo end and an in vitro end, the uterine assembly configured to be slidably housed in the main channel of the main assembly in such a way that the in vivo end of the uterine assembly is extendible outwardly away from the in vivo end of the main assembly, the uterine assembly including: a first negative pressure port, the first negative pressure port formed at a distal end of the in vivo end of the uterine assembly, the first negative pressure port configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the first negative pressure port being a state in which the first negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a second negative pressure, the non-homeostasis state of the first negative pressure port being a state in which the first negative pressure port does not apply a negative pressure

having a magnitude that is greater than or equal to the second negative pressure; a uterine expandable member, the uterine expandable member formed on the elongated body of the uterine assembly and adjacent to the first negative pressure port, the uterine expandable member configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the uterine expandable member being a state in which the uterine expandable member has a third overall volume, the non-homeostasis state of the uterine expandable member being a state in which the uterine expandable member has a fourth overall volume, the third overall volume greater than the fourth overall volume; and a second negative pressure port, the second negative pressure port formed adjacent to the uterine expandable member in such a way that the uterine expandable member is positioned between the first and second negative pressure ports, the second negative pressure port configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the second negative pressure port being a state in which the second negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a third negative pressure, the non-homeostasis state of the second negative pressure port being a state in which the second negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the third negative pressure, wherein the uterine assembly is configured to apply the second and third negative pressures in such a way that a combination of the second and third negative pressures is an amount of negative pressure required to collapse at least a portion of the uterine cavity towards the elongated body of the uterine assembly.
2. The hysteroscopic system of claim 1, wherein at least one of the following apply: the contact wall of the cervical seal member includes at least a portion that is formed in a concave shape; and/or the non-contact wall of the cervical seal member includes at least a portion that is formed in a convex shape; and/or the cervical seal member is formed along a plane that is substantially orthogonal to the first central axis; and/or the cervical seal member includes an outermost edge portion formed between the contact and non-contact walls of the cervical seal member, the outermost edge portion of the cervical seal member formed along a plane that is substantially orthogonal to the first central axis; and/or the contact wall of the cervical seal member includes a circular contact portion, the circular contact portion of the cervical seal member formed along a plane that is substantially orthogonal to the first central axis.
3. The hysteroscopic system of claim 1, wherein the contact wall of the cervical seal member is formed having at least one flexible portion; wherein the at least one flexible portion of the contact wall of the cervical seal member is configured to dynamically adapt to a surface topology of the cervix of the patient.
4. The hysteroscopic system of claim 1, wherein the cervical seal member includes a sealable member formed at a center of the cervical seal member; wherein the sealable member of the cervical seal member is coaxial to the first central axis; and wherein the sealable member of the cervical seal member is configured to allow the in vivo end of the uterine assembly to slide through in both directions; wherein the sealable member of the cervical seal member is configured to provide a hermetic seal around the uterine assembly when the uterine assembly is provided through the sealable member of the cervical seal member; and wherein the sealable member of the cervical seal member is configured to hermetically seal itself when the uterine assembly is not provided through the sealable member of the cervical seal member.
5. The hysteroscopic system of claim 1, wherein at least one of the following apply: wherein the cervical seal member is formed with sufficient rigidity that, when the third anchoring member applies the first negative pressure to collapse the at least one portion of the vaginal cavity towards the elongated body of the main assembly, the cervical seal member does not collapse or deform towards the third anchoring member; and/or wherein, when the first anchoring member is transitioned to the anchoring state such that the first anchoring member has the first overall volume, the first anchoring member has sufficient rigidity so as to not collapse or deform towards the third anchoring member when the third anchoring member applies the first negative pressure to collapse

the at least one portion of the vaginal cavity towards the elongated body of the main assembly; and/or wherein, when the second anchoring member is transitioned to the anchoring state such that the second anchoring member has the third overall volume, the second anchoring member has sufficient rigidity so as to not collapse or deform towards the third anchoring member when the third anchoring member applies the first negative pressure to collapse the at least one portion of the vaginal cavity towards the elongated body of the main assembly.

6. The hysteroscopic system of claim 1, wherein at least one of the following apply: wherein at least a portion of the in vivo end of the uterine assembly is configured to bend in one or more directions; and/or wherein at least a portion of the in vivo end of the uterine assembly is configured to extend and contract so as to change a length of the in vivo end of the uterine assembly.

7. The hysteroscopic system of claim 1, further comprising: a manipulator assembly, the manipulator assembly formed as an elongated body having an in vivo end and an in vitro end, the manipulator assembly configured to be slidably housed in the main channel of the main assembly when the uterine assembly is not housed in the main channel of the main assembly, the manipulator assembly including: an end-effector, the end-effector provided at a distal most end of the in vivo end of the manipulator assembly, the end-effector configurable to perform a surgical action.

8. The hysteroscopic system of claim 7, wherein at least one of the following apply: wherein at least a portion of the in vivo end of the manipulator assembly is configured to bend in one or more directions; and/or wherein at least a portion of the in vivo end of the manipulator assembly is configured to extend and contract so as to change a length of the in vivo end of the manipulator assembly.

9. The hysteroscopic system of claim 1, further comprising: a locking member, the locking member configured to transition between a locking state and an unlocking state such that, when in the locking state, the locking member is configured to lock movement of the uterine assembly relative to the main assembly when the uterine assembly is inserted through the main channel of the main assembly.

10. The hysteroscopic system of claim 9, wherein the locking member includes: a main locking body, the main locking body rotatable in one or more directions so as to transition between the unlocking state and the locking state; and/or a locking actuator, the locking actuator configured to transition between the locking state and the unlocking state when the main locking body is rotated.

11. The hysteroscopic system of claim 1, further comprising a controller, the controller configurable to perform at least one of the following: adjust a position of the main assembly in such a way that the contact wall of the cervical seal member is positioned to be in contact with the at least one portion of the cervix of the patient; and/or transition the first anchoring member between the anchoring state and the non-anchoring state; and/or transition the second anchoring member between the anchoring state and the non-anchoring state; and/or transition the third anchoring member between the anchoring state and the non-anchoring state; and/or transition the first negative pressure port between the homeostasis state and the non-homeostasis state; and/or transition the uterine expandable member between the homeostasis state and the non-homeostasis state; and/or transition the second negative pressure port between the homeostasis state and the non-homeostasis state.

12. A hysteroscopic system for managing post-partum hemorrhaging, the hysteroscopic system comprising: a main assembly, the main assembly formed as an elongated body having an in vivo end and an in vitro end, the in vivo end of the main assembly configured to be inserted into and housed in a vaginal cavity of a patient, the in vitro end of the main assembly configured to remain outside of the vaginal cavity of the patient, the main assembly including: a main channel formed through the elongated body of the main assembly between the in vivo and in vitro ends of the main assembly, the main channel formed along a first central axis; a first anchoring member, the first anchoring member formed on the elongated body of the main assembly, the first anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state

of the first anchoring member being a state in which the first anchoring member is expanded away from the elongated body of the main assembly; a second anchoring member, the second anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state of the second anchoring member being a state in which the second anchoring member is expanded away from the elongated body of the main assembly; and a vaginal seal member, the vaginal seal member formed at the in vitro end of the main assembly and having a central axis coaxial to the first central axis, the vaginal seal member having a contact wall and a non-contact wall opposite to the contact wall, the contact wall of the vaginal seal member configured to contact with at least a portion of a vulva of the patient, the vaginal seal member configured to hermetically isolate the vaginal cavity of the patient when: the contact wall of the vaginal seal member is positioned to be in contact with at least a portion of the vulva of the patient; and a third anchoring member, the third anchoring member formed between the first and second anchoring members, the third anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state of the third anchoring member being a state in which the third anchoring member applies at least a first negative pressure; and a uterine assembly, the uterine assembly formed as an elongated body having an in vivo end and an in vitro end, the uterine assembly configured to be slidably housed in the main channel of the main assembly, the uterine assembly including: a first negative pressure port, the first negative pressure port formed at a distal end of the in vivo end of the uterine assembly, the first negative pressure port configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the first negative pressure port being a state in which the first negative pressure port applies at least a second negative pressure; a uterine expandable member, the uterine expandable member formed on the elongated body of the uterine assembly and adjacent to the first negative pressure port, the uterine expandable member configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the uterine expandable member being a state in which the uterine expandable member is expanded away from the elongated body of the uterine assembly; and a second negative pressure port, the second negative pressure port formed adjacent to the uterine expandable member in such a way that the uterine expandable member is positioned between the first and second negative pressure ports, the second negative pressure port configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the second negative pressure port being a state in which the second negative pressure port applies at least a third negative pressure.

13. The hysteroscopic system of claim 12, wherein at least one of the following apply: the contact wall of the vaginal seal member includes at least a portion that is formed in a concave shape; and/or the non-contact wall of the vaginal seal member includes at least a portion that is formed in a convex shape; and/or the vaginal seal member is formed along a plane that is substantially orthogonal to the first central axis; and/or the vaginal seal member includes an outermost edge portion formed between the contact and non-contact walls of the vaginal seal member, the outermost edge portion of the vaginal seal member formed along a plane that is substantially orthogonal to the first central axis; and/or the contact wall of the vaginal seal member includes a circular contact portion, the circular contact portion of the vaginal seal member formed along a plane that is substantially orthogonal to the first central axis.

14. The hysteroscopic system of claim 12, wherein the contact wall of the vaginal seal member is formed having at least one flexible portion; wherein the at least one flexible portion of the contact wall of the vaginal seal member is configured to dynamically adapt to a surface topology of an area surrounding the vulva of the patient.

15. The hysteroscopic system of claim 12, wherein the vaginal seal member includes an opening formed at a center of the vaginal seal member; wherein the opening of the vaginal seal member is coaxial to the first central axis; wherein the opening of the vaginal seal member is configured to allow the in vivo end of the uterine assembly to slide through in both directions; wherein the

opening of the vaginal seal member is configured to provide a hermetic seal around the uterine assembly when the uterine assembly is provided through the opening of the vaginal seal member; and wherein the opening of the vaginal seal member is configured to hermetically seal itself when the uterine assembly is not provided through the opening of the vaginal seal member.

16. The hysteroscopic system of claim 12, wherein at least one of the following apply: wherein at least a portion of the in vivo end of the uterine assembly is configured to bend in one or more directions; wherein at least a portion of the in vivo end of the uterine assembly is configured to extend and contract so as to change a length of the in vivo end of the uterine assembly; wherein the uterine assembly is configured to apply the second and third negative pressures in such a way that a combination of the second and third negative pressures is an amount of negative pressure required to collapse at least a portion of the uterine cavity towards the elongated body of the uterine assembly.

17. The hysteroscopic system of claim 12, further comprising: a manipulator assembly, the manipulator assembly formed as an elongated body having an in vivo end and an in vitro end, the manipulator assembly configured to be slidably housed in the main channel of the main assembly when the uterine assembly is not housed in the main channel of the main assembly, the manipulator assembly including: an end-effector, the end-effector provided at a distal most end of the in vivo end of the manipulator assembly, the end-effector configurable to perform a surgical action.

18. The hysteroscopic system of claim 17, wherein at least a portion of the in vivo end of the manipulator assembly is configured to bend in one or more directions.

19. The hysteroscopic system of claim 17, wherein at least a portion of the in vivo end of the manipulator assembly is configured to extend and contract so as to change a length of the in vivo end of the manipulator assembly.

20. The hysteroscopic system of claim 12, further comprising a controller, the controller configurable to perform at least one of the following: adjust a position of the main assembly in such a way that the contact wall of the vaginal seal member is positioned to be in contact with the at least one portion of the vulva of the patient; and/or transition the first anchoring member between the anchoring state and the non-anchoring state; and/or transition the second anchoring member between the anchoring state and the non-anchoring state; and/or transition the third anchoring member between the anchoring state and the non-anchoring state; and/or transition the first negative pressure port between the homeostasis state and the non-homeostasis state; and/or transition the uterine expandable member between the homeostasis state and the non-homeostasis state; and/or transition the second negative pressure port between the homeostasis state and the non-homeostasis state; and/or apply the second and third negative pressures in such a way that a combination of the second and third negative pressures is an amount of negative pressure required to collapse at least a portion of the uterine cavity towards the elongated body of the uterine assembly.

21. A hysteroscopic system for managing post-partum hemorrhaging, the hysteroscopic system comprising: a main assembly, the main assembly formed as an elongated body having an in vivo end and an in vitro end, wherein, when in operation, the in vivo end of the main assembly is configured to be inserted into and housed in a vaginal cavity of a patient and the in vitro end of the main assembly is configured to remain outside of the vaginal cavity of the patient, the main assembly including: a main channel formed through the elongated body of the main assembly between the in vivo and in vitro ends of the main assembly, the main channel formed along a first central axis; a cervical seal member, the cervical seal member formed at the in vivo end of the main assembly and having a central axis coaxial to the first central axis, the cervical seal member having a contact wall and a non-contact wall opposite to the contact wall of the cervical seal member, the contact wall of the cervical seal member configured to contact with at least a portion of a cervix of the patient, the cervical seal member configured to hermetically isolate a uterine cavity of the patient from the vaginal cavity of the patient when: the contact wall of the cervical seal member is

positioned to be in contact with at least a portion of the cervix of the patient; a first anchoring member, the first anchoring member formed on the elongated body of the main assembly and adjacent to the cervical seal member, the first anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state of the first anchoring member being a state in which the first anchoring member has a first overall volume, the non-anchoring state of the first anchoring member being a state in which the first anchoring member has a second overall volume, the first overall volume greater than the second overall volume; a second anchoring member, the second anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state of the second anchoring member being a state in which the second anchoring member has a third overall volume, the non-anchoring state of the second anchoring member being a state in which the second anchoring member has a fourth overall volume, the third overall volume greater than the fourth overall volume; a third anchoring member, the third anchoring member formed between the first and second anchoring members, the third anchoring member configured to transition between an anchoring state and a non-anchoring state, the anchoring state of the third anchoring member being a state in which the third anchoring member applies at least a first negative pressure, the first negative pressure being an amount of negative pressure required to collapse at least a portion of the vaginal cavity towards the elongated body of the main assembly, the non-anchoring state of the third anchoring member being a state in which the third anchoring member does not apply at least the first negative pressure; and a vaginal seal member, the vaginal seal member formed at the in vitro end of the main assembly and having a central axis coaxial to the first central axis, the vaginal seal member having a contact wall and a non-contact wall opposite to the contact wall, the contact wall of the vaginal seal member configured to contact with at least a portion of a vulva of the patient, the vaginal seal member configured to hermetically isolate the vaginal cavity of the patient when: the contact wall of the vaginal seal member is positioned to be in contact with at least a portion of the vulva of the patient; and a uterine assembly, the uterine assembly formed as an elongated body having an in vivo end and an in vitro end, the uterine assembly configured to be slidably housed in the main channel of the main assembly in such a way that the in vivo end of the uterine assembly is extendible outwardly away from the in vivo end of the main assembly, the uterine assembly including: a first negative pressure port, the first negative pressure port formed at a distal end of the in vivo end of the uterine assembly, the first negative pressure port configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the first negative pressure port being a state in which the first negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a second negative pressure, the non-homeostasis state of the first negative pressure port being a state in which the first negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the second negative pressure; a uterine expandable member, the uterine expandable member formed on the elongated body of the uterine assembly and adjacent to the first negative pressure port, the uterine expandable member configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the uterine expandable member being a state in which the uterine expandable member has a third overall volume, the non-homeostasis state of the uterine expandable member being a state in which the uterine expandable member has a fourth overall volume, the third overall volume greater than the fourth overall volume; and a second negative pressure port, the second negative pressure port formed adjacent to the uterine expandable member in such a way that the uterine expandable member is positioned between the first and second negative pressure ports, the second negative pressure port configured to transition between a homeostasis state and a non-homeostasis state, the homeostasis state of the second negative pressure port being a state in which the second negative pressure port applies a negative pressure having a magnitude that is greater than or equal to a third negative pressure, the non-homeostasis state of the second negative pressure port being a state in which the second negative pressure port does not apply a negative pressure having a magnitude that is greater than or equal to the third negative pressure.

22. The hysteroscopic system of claim 21, wherein at least one of the following apply: the contact wall of the cervical seal member includes at least a portion that is formed in a concave shape; and/or the non-contact wall of the cervical seal member includes at least a portion that is formed in a convex shape; and/or the contact wall of the vaginal seal member includes at least a portion that is formed in a concave shape; and/or the non-contact wall of the vaginal seal member includes at least a portion that is formed in a convex shape; and/or the cervical seal member is formed along a plane that is substantially orthogonal to the first central axis; and/or the cervical seal member includes an outermost edge portion formed between the contact and non-contact walls of the cervical seal member, the outermost edge portion of the cervical seal member formed along a plane that is substantially orthogonal to the first central axis; and/or the contact wall of the cervical seal member includes a circular contact portion, the circular contact portion of the cervical seal member formed along a plane that is substantially orthogonal to the first central axis; and/or the vaginal seal member is formed along a plane that is substantially orthogonal to the first central axis; and/or the vaginal seal member includes an outermost edge portion formed between the contact and non-contact walls of the vaginal seal member, the outermost edge portion of the vaginal seal member formed along a plane that is substantially orthogonal to the first central axis; and/or the contact wall of the vaginal seal member includes a circular contact portion, the circular contact portion of the vaginal seal member formed along a plane that is substantially orthogonal to the first central axis.

23. The hysteroscopic system of claim 21, wherein the contact wall of the cervical seal member is formed having at least one flexible portion; wherein the at least one flexible portion of the contact wall of the cervical seal member is configured to dynamically adapt to a surface topology of the cervix of the patient; wherein the contact wall of the vaginal seal member is formed having at least one flexible portion; wherein the at least one flexible portion of the contact wall of the vaginal seal member is configured to dynamically adapt to a surface topology of an area surrounding the vulva of the patient.

24. The hysteroscopic system of claim 21, wherein the cervical seal member includes a sealable member formed at a center of the cervical seal member; wherein the sealable member of the cervical seal member is coaxial to the first central axis; wherein the sealable member of the cervical seal member is configured to allow the in vivo end of the uterine assembly to slide through in both directions; wherein the sealable member of the cervical seal member is configured to provide a hermetic seal around the uterine assembly when the uterine assembly is provided through the sealable member of the cervical seal member; wherein the sealable member of the cervical seal member is configured to hermetically seal itself when the uterine assembly is not provided through the sealable member of the cervical seal member.

25. The hysteroscopic system of claim 21, wherein the vaginal seal member includes an opening formed at a center of the vaginal seal member; wherein the opening of the vaginal seal member is coaxial to the first central axis; wherein the opening of the vaginal seal member is configured to allow the in vivo end of the uterine assembly to slide through in both directions; wherein the opening of the vaginal seal member is configured to provide a hermetic seal around the uterine assembly when the uterine assembly is provided through the opening of the vaginal seal member; and wherein the opening of the vaginal seal member is configured to hermetically seal itself when the uterine assembly is not provided through the opening of the vaginal seal member.

26. The hysteroscopic system of claim 21, wherein at least one of the following apply: wherein the cervical seal member is formed with sufficient rigidity that, when the third anchoring member applies the first negative pressure to collapse the at least one portion of the vaginal cavity towards the elongated body of the main assembly, the cervical seal member does not collapse or deform towards the third anchoring member; and/or wherein, when the first anchoring member is transitioned to the anchoring state such that the first anchoring member has the first overall volume, the first anchoring member has sufficient rigidity so as to not collapse or deform towards the third

anchoring member when the third anchoring member applies the first negative pressure to collapse the at least one portion of the vaginal cavity towards the elongated body of the main assembly; and/or wherein, when the second anchoring member is transitioned to the anchoring state such that the second anchoring member has the third overall volume, the second anchoring member has sufficient rigidity so as to not collapse or deform towards the third anchoring member when the third anchoring member applies the first negative pressure to collapse the at least one portion of the vaginal cavity towards the elongated body of the main assembly; and/or wherein at least a portion of the in vivo end of the uterine assembly is configured to bend in one or more directions; and/or wherein at least a portion of the in vivo end of the uterine assembly is configured to extend and contract so as to change a length of the in vivo end of the uterine assembly.

27. The hysteroscopic system of claim 21, further comprising: a manipulator assembly, the manipulator assembly formed as an elongated body having an in vivo end and an in vitro end, the manipulator assembly configured to be slidably housed in the main channel of the main assembly when the uterine assembly is not housed in the main channel of the main assembly, the manipulator assembly including: an end-effector, the end-effector provided at a distal most end of the in vivo end of the manipulator assembly, the end-effector configurable to perform a surgical action.

28. The hysteroscopic system of claim 27, wherein at least one of the following apply: wherein at least a portion of the in vivo end of the manipulator assembly is configured to bend in one or more directions; and/or wherein at least a portion of the in vivo end of the manipulator assembly is configured to extend and contract so as to change a length of the in vivo end of the manipulator assembly.

29. The hysteroscopic system of claim 21, further comprising a controller, the controller configurable to perform at least one of the following: adjust a position of the main assembly in such a way that the contact wall of the cervical seal member is positioned to be in contact with the at least one portion of the cervix of the patient; and/or transition the first anchoring member between the anchoring state and the non-anchoring state; and/or transition the second anchoring member between the anchoring state and the non-anchoring state; and/or transition the third anchoring member between the anchoring state and the non-anchoring state; and/or transition the first negative pressure port between the homeostasis state and the non-homeostasis state; and/or transition the uterine expandable member between the homeostasis state and the non-homeostasis state; and/or transition the second negative pressure port between the homeostasis state and the non-homeostasis state; and/or apply the second and third negative pressures in such a way that a combination of the second and third negative pressures is an amount of negative pressure required to collapse at least a portion of the uterine cavity towards the elongated body of the uterine assembly.